



**Verified Carbon
Standard**

LICHUAN CITY DOMESTIC WASTE LANDFILL SITE BIOGAS POLLUTION TREATMENT AND COMPREHENSIVE UTILIZATION PROJECT



New Power
百川畅银

Project title	Lichuan City Domestic Waste Landfill Site Biogas Pollution Treatment and Comprehensive Utilization Project
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Crediting period	05-September-2024 to 04-September-2034 (both days included)
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Lichuan City Domestic Waste Landfill Site Biogas Pollution Treatment and Comprehensive Utilization Project (hereafter referred to as the project) is located in No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China. The project is invested, constructed and operated by Lichuan City BCCY New Energy Co., Ltd. The objective is to collect the landfill gas (LFG) from Lichuan city domestic waste landfill site (hereafter referred to as Lichuan landfill site) and utilize it to generate electricity.

Lichuan landfill site is located at No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City. It started operation on December, 2007 with a total design effective capacity of 1,690,000 m³ and will cease on 2026¹.

The project has a total designed capacity 2MW (2*1000kW) and combusts the LFG, which contains nearly 50% of methane, to generate electricity and export it to Central China Power Grid (CCPG). The project started construction on 20-April-2024 and started operation on 05-September-2024. The crediting period of the project started on 05-September-2024 and will end on 04-September-2034.

The project collects and utilizes LFG from Lichuan landfill site for power generation, which would otherwise be released into the atmosphere. Methane in the LFG has been emitted to the atmosphere since the landfill started operation. Therefore, the project avoids methane emissions and, at the same time, electricity is generated from a renewable energy source, instead of being provided by the grid.

Scenario existing prior to the start of the implementation of the project activity (the same as baseline scenario)

- LFG from Lichuan landfill site is released into the atmosphere.
- Equivalent electricity generated by the project is supplied by the CCPG, which is dominated by fossil fuel based power plants.

¹ According to the land certificate issued on 24-April-2020 by the government, the land is solely belonged to Lichuan City Environmental Sanitation Service Center. The purpose of this land is for public facilities. According to Article 12 of the "Provisional Regulations on the transfer and transfer of the right to use of state owned land in People's Republic of China", the term of this land use right is 50 years. Therefore, from 2026 to 2034, this land belongs to Lichuan City Environmental Sanitation Service Center. According to the venue lease contract signed with Lichuan City Environmental Sanitation Service Center on 18-January-2024, this project has the right to use the land located at the landfill site. The lease period is from January 18, 2024 to the end of the project's shutdown. Therefore, even though the landfill will be closed in 2026, if the project has not been shut down, and the land can still be utilized.

The total power generation during the operation period is estimated to be 75,670 MWh. The estimated average GHG emission reduction of the project is 45,669 tCO₂e and the total GHG emission reductions and removals are 456,696 tCO₂e during the fixed 10 years crediting period.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	05-January-2025 (the date when the FVR was completed)	VCS	LGAI Technological Center, S.A. (Applus+ Certification)	N/A

1.3 Sectoral Scope and Project Type

The project is a non-AFOLU project and not a grouped project activity.

Sectoral scope ²	1; 13
Project activity type	Energy industries (renewable-/non-renewable sources); Waste handling and disposal

The project is a LFG recovery to power project and not an AFOLU project. Therefore, the following table isn't applicable to the project.

Sectoral scope	N/A
AFOLU project category ³	N/A
Project activity type	N/A

1.4 Project Eligibility

1.4.1 General eligibility

As per section 2.1.1 of VCS Standard (version 4.7), the scope of the VCS Program includes:

VCS Standard Scope	Fulfilled (Yes/No)	Justification
(1) The seven Kyoto Protocol greenhouse gases.	Yes	The project is a LFG recovery to power project. Thus, the project is applicable to this scope.
(2) Ozone-depleting substances (ODS).	N/A	The project activity does not involve any Ozone depleting substances.

² Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

³ See Appendix 1 of the VCS Standard

(3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	N/A	The project activity is supported by the CDM methodology.
(4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Yes	The applied methodologies AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0) of the project are methodologies approved under CDM Program, which is a VCS approved GHG program.
(5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	N/A	This is not a Jurisdictional REDD+ program nor a nested REDD+ project, hence this eligibility criterion is not applicable.

As per section 2.1.3 of VCS Standard (version 4.7), the VCS Program also excludes the following project activities under the circumstances indicated in Table 1, below.

Excluded Activity	Fulfilled (Yes/No)	Justification
Grid-connected electricity generation activities using hydroelectric power plants.	N/A	The project is not involved in hydroelectric power plant. The project involves capturing LFG produced from Lichuan landfill site and use it to generate electricity which will be supplied to the grid.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants.	N/A	The project is not involved in wind, geothermal, or solar photovoltaic (PV) power plant. The project involves capturing LFG produced from Lichuan landfill site and use it to generate electricity which will be supplied to the grid.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration.	N/A	The project is not involved in recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The project involves

		capturing LFG produced from Lichuan landfill site and use it to generate electricity which will be supplied to the grid.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.	N/A	The project is not involved in generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The project involves capturing LFG produced from Lichuan landfill site and use it to generate electricity which will be supplied to the grid.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel.	N/A	The project does not generate electricity and/or thermal energy using fossil fuels. On the contrary, the project collects and utilizes LFG to generate electricity. The project does not involve switching from a higher to a lower carbon content fossil fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).	N/A	The project does not include the replacement of electric lighting. The project involves capturing LFG produced from Lichuan landfill site and use it to generate electricity which will be supplied to the grid.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.	N/A	The project does not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers. The project involves capturing LFG produced from Lichuan landfill site and use it to generate electricity which will be supplied to the grid.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A	The project involves capturing LFG produced from Lichuan landfill site

		and use it to generate electricity which will be supplied to the grid.
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In conclusion, the project is included under the scope of the VCS Program and not excluded under Table 1 of the VCS Standard.

According to 3.1 of VCS standard (Version 4.7), establishing a consistent and standardized certification process is critical to ensuring the integrity of VCS projects. Accordingly, certain high-level requirements must be met by all projects, as set out below.

Reference to VCS Standard (Version 4.7)	Requirements	Justification
3.1.1	Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1.	The project meets all applicable rules and requirements set out under the VCS Program, including this document.
3.1.2	Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1. The list of methodologies and their validity periods is available on the Verra website.	The project applies the methodologies AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0) along with tools or modules as applicable developed under the United Nations Clean Development Mechanism, which can be used for projects and programs registering with VCS as per requirements available on the Verra website.
3.1.3	Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.22 below. Projects shall update to the latest version of the methodology when	The project applies the methodologies AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0), which are the latest version.

	reassessing the baseline or renewing a crediting period.	
3.1.4	Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced.	According to the environment impact assessment report (EIA) of the project, the project complies with all Chinese relevant laws and regulations.
3.1.5	Where projects apply methodologies that permit the project proponent its own choice of model (see the VCS Program Definitions for the definition of model), the model shall meet the requirements set out in the VCS Methodology Requirements, and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or carbon dioxide removals).	Not applicable as there is no model needs to be chosen by the project proponent as per the applied methodologies AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0).
3.1.6	Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet the requirements set out in the VCS Methodology Requirements.	Not applicable as there are no third-party default factor or standard need to be chosen by the project proponent as per the applied methodologies AMS-III.G. (Version 10.0) and AMS-I.D. (Version 18.0).

3.1.7	Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.	Not applicable. The project obeys the rules and requirements of the VCS Program
3.1.8	Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits (see the VCS Program Definitions for the definition of capacity limit) and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	According to AMS-III.G. (Version 10.0), the project results in aggregate emission reduction of less than 60kt CO ₂ equivalent annually from all Type III components. The project is a small-scaled project. The project is not a fragmented part of a larger project or activity.
3.1.9	Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 Document History and Effective Dates or equivalent for other program documents, and are listed in a companion Summary of Effective Dates document which corresponds with each update.	The project has followed the latest requirements of Verra.

According to Section 3.8.1 of VCS Standard (version 4.7), Non-AFOLU projects shall complete validation within two years of the project start date. The project is a non-AFOLU project, and it requested for pipeline listing on Verra Website in October, 2024. The project status is Under validation (Public comment period closed) now. The project start date is 05-September-2024 and will complete validation before 04-September-2026, which is within two years of the project start

date. The project meets requirements related to the pipeline listing deadline and the validation deadline.

According to Section 4.1.5 of VCS Standard (version 4.7), the project is listed on the project pipeline with a status of under validation before the opening meeting with the validation/verification body, such opening meeting representing the beginning of the validation process. Further, validation shall not begin until the 30-day public comment period has begun. The 30-day public comment period began on 18-November-2024, and the opening meeting began on 30-December-2024. Therefore, the project meets the requirement related to the opening meeting with the validation/verification body.

1.4.2 AFOLU project eligibility

The project is not an AFOLU project. Thus, this section is not applicable.

1.4.3 Transfer project eligibility

The project is not a transfer project. Thus, this section is not applicable.

1.5 Project Design

The project has been designed as:

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

The project is not a grouped project. Thus, this section is not applicable.

1.6 Project Proponent

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Minister of carbon emission department
Address	10/F, Kailin Building, 99 Ruyi West Road, Zhengzhou, China
Telephone	+86 371 56737901
Email	lwang@bccynewpower.com

1.7 Other Entities Involved in the Project

Organization name	Lichuan City BCCY New Energy Co., Ltd.
Role in the project	Project Operation Entity
Contact person	Yangyang Zhang
Title	Project Manager
Address	10/F, Kailin Building, 99 Ruyi West Road, Zhengzhou, China
Telephone	+86 371 65521780
Email	zhangyangyang@bccynewpower.com

1.8 Ownership

Lichuan City BCCY New Energy Co., Ltd. is the sole project owner of the project. The Project Filing Certificate and approval of Environmental Impact Assessment (EIA) and the Business License of the project owner are the evidences for legislative right. Lichuan City BCCY New Energy Co., Ltd. is a wholly-owned subsidiary of Henan BCCY Environmental Energy Co., Ltd., which is responsible for the acquisition of approvals, construction and operation of project. It obeys the management of Henan BCCY Environmental Energy Co., Ltd., including application for VCS carbon emission reduction project. Therefore, the project proponent is Henan BCCY Environmental Energy Co., Ltd.

1.9 Project Start Date

Project start date	05-September-2024
Justification	According to VCS standard, the project start date is the date on which activities that lead to the generation of GHG emission reductions or removals are implemented. The project started operation on 05-September-2024. Thus, this date is the start date of the project.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other
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	According to VCS standard (version 4.7), the total project crediting period shall be either seven years (twice renewable for a total of up to 21 years) or ten years fixed, except for AFOLU and GCS projects as defined below and in the GCS Requirements respectively. The project is a landfill gas power generation project, ten years fixed crediting period is applied.
Start and end date of first or fixed crediting period	05-September-2024 to 04-September-2034

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals (ERRs) of the project:

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
05-September-2024 to 31-December-2024	17,847
01-January-2025 to 31-December-2025	59,323
01-January-2026 to 31-December-2026	63,430 ⁴
01-January-2027 to 31-December-2027	56,543
01-January-2028 to 31-December-2028	50,640
01-January-2029 to 31-December-2029	45,566
01-January-2030 to 31-December-2030	41,188
01-January-2031 to 31-December-2031	37,398
01-January-2032 to 31-December-2032	34,104
01-January-2033 to 31-December-2033	31,229
01-January-2034 to 04-September-2034	19,428
Total estimated ERRs during the first or fixed crediting period	456,696

⁴ According to paragraph 4 of AMS-III.G. (version 10.0), emission reductions are required less than or equal to 60 kt CO₂ equivalent annually from all Type III components of the project activity. In this project, the emission reduction from methane destroy is 57,405 tCO₂ in 2026, which is less than 60 kt CO₂. 63430 = 57,405 + 6,025. Therefore, small scale meth is applicable to the project.

Total number of years	10
Average annual ERRs	45,669

1.12 Description of the Project Activity

GHG emission reductions will be achieved through combustion of the recovered methane gas by gas engines, which would be otherwise emitted to the atmosphere, and the generation of electricity from the landfill gas, which is supplied by Central China Power Grid (CCPG) prior to the implementation of the project.

Description of the technology in the project

The project consists in LFG collection, transmission and treatment system, with subsequent electricity generation and grid connection system.

LFG collection system

LFG is extracted from MSW under negative pressure by blower pumps and moved through wells, then collected by gas collection stations and transferred by sub-pipes and a main pipe to LFG treatment equipment. Flow rate of the LFG is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

LFG treatment system

Prior to electricity generation, LFG is treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.

Electricity generation and grid connection system

2 gas engines of 1000kW rated power each are fed with the LFG and generate electricity, which is then exported to the grid.

The project does not include any flare system. The project uses recovered landfill gas to generate electricity with a destruction efficiency of 100 percent in the gas generator. During the operation period, the LFG flow can be adjusted based on LFG demand of the project. Even in case all the gas generators have to stop operation for emergency and the landfill site is able to hold LFG. The solenoid valve on the intake pipeline at the front end of each gas generator will also be automatically activated. The LFG will be temporarily stored in the landfill site and pipelines and impossible pass into the atmosphere. When the engines are scheduled to shut down, the LFG collection system reduces the gas in advance. There was no LFG collected and left in the entire project system during the planned shutdown. At this point in time, LFG did not enter through the roots blower pumps, LFG was stored in the pipeline between the landfill and the pre-treatment facility, the landfill was equipped with gas wells that have the capacity to hold the amount of LFG generated. At the beginning of the design process, the design took into account the length of the

pipeline and the safety features, and the pipeline has the capacity to store the LFG. Safety has been taken into account during the design and commissioning stages of the project and the design is compliant. When the unit is shut down in an emergency, intake pressure is increased and the roots blower pumps will reduce its frequency and extract less LFG from the landfill site. It is a type of interlocking control, ensuring that the pressure in the unit is balanced and that a very small portion of the LFG stored in the pipework can be accommodated. At the same time, the maintenance personnel will handle the malfunction as soon as possible. So, the LFG won't pass into the atmosphere. After the engines return to normal, the LFG collection system will adjust to increase the intake volume.

Table 1.1 Equipment technical parameters⁵

Main Equipment	Parameter	Value
LFG collection system ⁶	Type	An integrated system consisting of gas collection wells and pipelines
LFG treatment system	Type	LFG-1500
	Manufacturer	Nanjing Carbon Ring Biomass Energy Co., Ltd.
	Handle gas volume	1500 Nm ³ /h
Rotary air compressor	Type	3L42WC
	Number	2
	Manufacturer	NANTONG HENGRONG BLOWER AND PUMP CO., LTD
	Flow volume	17.18 m ³ /min
	Motor power	22 kW
	Speed	1450 r/min
	Type	OL52WC
	Number	1
	Manufacturer	NANTONG JULONG FLUID MACHINERY CO., LTD
	Flow volume	32.8 m ³ /min

⁵ The data are from nameplate and technical agreement.

⁶ Since the biogas collection system is a system composed of gas collection wells and pipelines, there are no specific equipment parameters.

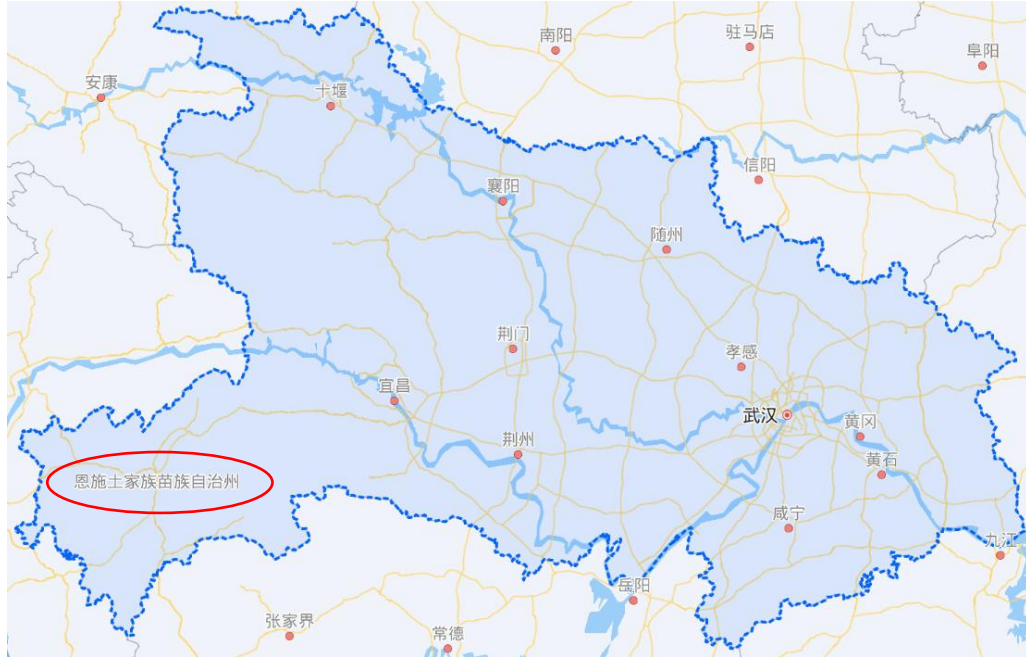
Generator set	Motor power	37 kW
	Speed	1450 r/min
	Type	1000GF-NK
	Manufacturer	JINAN DIESEL ENGINE CO., LTD
	Rated capacity	1000 kW
	Rated rotation speed	1000 r/min
	Capacity factor	0.8 (lagging)
	Lifetime	15 years

1.13 Project Location

The project is located in Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China. The geographical coordinates of the project site are longitude 108°58'53.205" E and latitude 30°18'1.403" N. Figure 1.1 shows the location of the Project.



Map of People's Republic of China



Map of Hubei Province



the location of the project site

Figure 1.1 Geographical locations of the project

1.14 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity (the same as baseline scenario)

- LFG from Lichuan landfill site is released into the atmosphere.

- Equivalent electricity generated by the project is supplied by Central China Power Grid, which is dominated by fossil fuel based power plants.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Chinese relevant laws and regulations. Mainly include:

1. Renewable Energy Law of the People's Republic of China;
2. National Action plan for the collection and Utilization of municipal landfill gas;
3. Law of the People's Republic of China on the Prevention and Control of Environmental Pollution by Solid Waste;
4. Catalogue for the Guidance of Industrial Structure Adjustment (2011 version);
5. "14th Five-Year" Plan for Energy Development of Hubei Province

The project obtained the approval letters from local governmental authorities: Investment Supervisory Department, as well as Environment Protection Agency (EPA). The two approvals well demonstrate that local governments permit the construction of the project. Hence, the project is in compliance with laws, status and other regulatory frameworks.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The project is located in Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China. The project captures the landfill gas and generates electricity, which is fed into Central China Power Grid, displacing energy that would otherwise be generated by fossil fuels.

The primary objective of the project is to avoid greenhouse gases emission through landfill gas capture, purification, and injection in a grid connection system, while contributing to the environmental, social, and economic sustainability by minimizing global climate changes and local air pollution. Additional benefits include the increase of employment opportunities through full-time and permanent positions, etc.

The Project will contribute to sustainable development in the following ways:

- **SDG 13 Climate Action:** The project will recover and destroy landfill gas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to the China's commitment to peak carbon dioxide emissions before 2030;

- **SDG 7 Affordable and Clean Energy:** Increase electricity production from renewable sources, which will certainly reduce the consumption of fossil fuel in Central China Power Grid and further help to achieve the national action to promote renewable energy development;
- **SDG 8 Decent Work and Economic Growth:** The project will provide new jobs opportunities and increase tax revenue, which will have a positive effect on the local economy.

Table1.2: Sustainable Development Contributions

SDG Target	SDG Indicator	Project activity contribution
13.0	Tonnes of greenhouse gas emissions avoided or removed	The project will avoid anthropogenic emissions of greenhouse gases (GHG) of 456,696 tCO ₂ e during the crediting period (05-September-2024 to 04-September-2034).
8.5	Total number of job opportunities	The project activity employs 8 permanent and full-time employees through its operational company Lichuan City BCCY New Energy Co., Ltd. during the crediting period (05-September-2024 to 04-September-2034).
7.2	MWh of renewable energy generated	Total 75,670 MWh of renewable energy will be generated during the crediting period (05-September-2024 to 04-September-2034).

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not applicable as not AFOLU project.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further information available.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The purpose of the project is to utilize LFG to produce electricity, which is supplied to CCPG. The entire project must adhere to local regulations, rules, and obtain necessary permissions. The following steps were adapted to identify stakeholders (based on Section 3.18 of VCS Standard—version 4.7). Step 1: List all individuals or groups who may have an impact on or be affected by the project. Step 2: Conduct a well-being ranking of local or community stakeholders. Step 3: Analyze the interests, motivations for participation, and relationships with other stakeholders of each stakeholder group. Step 4: Analyze the influence and importance of each potential stakeholder group. Step 5: Publish the results of the identified stakeholders. If anyone else believes they are one of the project's stakeholders, they can directly contact the project supporters for consultation. The final stakeholders of this project are as followed: 1. local residents 2. local authorities 3. employees of Lichuan landfill site 4. employees of Lichuan City BCCY New Energy Co., Ltd. 5. the project proponent
Legal or customary tenure/access rights	The project is located at Lichuan landfill site and does not involve the land issues.

	According to the project cooperation agreement signed between the project proponent and Lichuan City Urban Management Law Enforcement Bureau (the supervising authority of Lichuan landfill site), the project proponent has the right to use their LFG for power generation. According to the venue lease contract signed with Lichuan City Environmental Sanitation Service Center⁷ (the operation entity of Lichuan landfill site), the project has the right to use the land located at the landfill site.
Stakeholder diversity and changes over time	Stakeholders identified for this project activity are diverse in nature – in terms of socio-economic status, age and gender wise and the diversity is unlikely to change over time.
Expected changes in well-being	1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected; 2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 75,670 MWh, which could replace the equivalent electricity from fossil fuel based grid.
Location of stakeholders	The location of stakeholders is Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China.
Location of resources	The location of resources is at Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	19-April-2024
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⁷ Lichuan City Environmental Sanitation Service Center is a subordinate unit of Lichuan City Urban Management Law Enforcement Bureau, and is under its supervision.

Stakeholder engagement process	The stakeholders were informed about the stakeholder meeting through posters on 18-March-2024. In the bulletin, the company invited the potential stakeholders to get to know their opinions and/or suggestions about the implementation of the project and VCS application of the project. The stakeholder consultancy meeting for the parties interested in the project was organized at Lichuan landfill site on 19-April-2024 to collect opinions from all stakeholders, such as representatives of local residents and local authorities, Lichuan landfill site, the project proponent and so on. The project proponent appointed one person to make a meeting minute. Bulletin and meeting are in Chinese. The stakeholder meeting was attended by 13 people, six of whom were women.
Consultation outcome	During the stakeholder consultancy meeting, some stakeholders worried that the project will bring noise, land occupying and pollution on employees and local residents' living. For these issues, the project proponent will offer some measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents. All attendances to the meeting were satisfied to the measures. In general, all participants have positive perception on project design and implementation. The project would actually facilitate the development of the local economy and increase the income of local residents. During the stakeholder consultancy meeting, the project proponent assured everyone that they would comply with labor laws and ensure the legitimate rights and interests of workers. The project proponent also introduced the VCS validation and verification process to everyone.
Ongoing communication	Expect for introducing the project information, ongoing communication and grievance mechanism were also explained during the local stakeholder consultation meeting. The project proponent will regularly hold stakeholder meetings to exchange project progress, issues, and solutions with local residents. For employees of the enterprise, the project proponent establishes a labor union organization and holds at least one employee meeting every year to pay attention to the needs of each employee and

	collect their opinions to better operate the project and improve employee happiness. Meanwhile, a grievance expression book will be put in the office of the project, which is used to collect all stakeholders' views about the project. Project proponent will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Project Participant will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible. Until now, no comments or Input / Grievance Expression received from stakeholder.
Stakeholder input	The project did not receive major negative input during the consultation. The project design need not to have any updates. All stakeholders were pleased with the development of the project. The project would actually facilitate the development of the local economy and increase the income of local residents.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Based on the principle of friendly consultation and on the basis of truly and fully expressing their respective wishes, the project proponent and Lichuan City Urban Management Law Enforcement Bureau signed an cooperation agreement on this project. The feasibility study report of the project was approved by the Development and Reform Bureau of Lichuan City on 27-February-2024. An environmental impact assessment (EIA) report of the project was approved by Enshi Tujia and Miao autonomous Prefecture Ecological Environment Bureau Lichuan City branch on 18-April-2024. According to stakeholder consultation meetings, local residents have expressed support for the project. In general, the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.
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	There are not ongoing or unresolved conflicts during the project validation period and monitoring period. Therefore, the project does not exacerbate nor influence the outcomes of unresolved conflicts.
Outcome of FPIC	The feasibility study report of the project was approved by the Development and Reform Bureau of Lichuan City on 27-February-2024. An environmental impact assessment (EIA) report of the project was approved by Enshi Tujia and Miao autonomous Prefecture Ecological Environment Bureau Lichuan City branch on 18-April-2024. The project is located at Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China. According to the venue lease contract signed with Lichuan City Environmental Sanitation Service Center (the operation entity of Lichuan landfill site) on 18-January-2024, the land used in this project is Lichuan landfill site, which does not involve land disputes. Based on questionnaire surveys and stakeholder consultation meetings, local residents have expressed support for the project. The project was constructed with the consent of government departments at all levels. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Development process	The written procedure in place for addressing stakeholder's grievances comprises the following steps: 1. Identification Complaints and suggestions will be classified according to their seriousness and potential impact on the project's objectives. Priority classification will determine the response deadlines and subsequent actions. 2. Responsibility for Response PP will be will be responsible for periodically reviewing the means available for stakeholder communication. After analyzing the complaints and suggestions presented in the media, the
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	responsible managers will coordinate the response activities, involving the relevant parties as necessary. 3. Investigation and Analysis The responsible team will conduct a detailed investigation to fully understand the nature of the complaint or suggestion. Potential impacts and possible solutions will be assessed. 4. Communication with the Stakeholder The stakeholder who submitted the complaint or suggestion will be kept informed of the progress of the investigation. Communication will be carried out in a clear and timely manner. 5. Resolution and Implementation of Improvements Based on the analysis, necessary corrective or preventive actions will be identified. The proposed improvements will be implemented in a coordinated manner. 6. Feedback and Learning After the resolution, feedback will be sought from the stakeholder to assess the effectiveness of the actions taken. The lessons learned will be documented in order to continuously improve the process.
Grievance redress procedure	If there are conflicts of interest and dissatisfaction among stakeholders, they can choose to directly appeal to the project proponent or initiate an appeal meeting, which is usually the most effective way to solve the problem. Stakeholders can file appeals or complaints during the meeting through the phone number of the relevant contact person assigned by the project proponent. In addition, the progress of the project is made public on the VCS Pipeline website, accepting public feedback. If stakeholders have appeals and complaints, they can directly provide feedback to Verra. Their opinions will be taken into consideration, and the project proponent will promptly respond and provide feedback to Verra. In terms of government regulation, sometimes complaint mechanisms at the government level can also serve as an effective solution. This project is supervised by the local Ecological Environment Bureau and Development and Reform Bureau. The above institutions all have public complaint telephone channels, through which stakeholders can provide feedback on their opinions.

The project proponent will respond to it and provide feedback on improvement to government regulatory agencies.

In the project, the project proponent has designated a staff member to record and collect dissatisfaction and complaints related to conflicts of interest, who plays an important role in handling these common conflicts and appeals. Once a report regarding the appeal is received, the project proponent will contact and discuss with relevant communities or other stakeholders within 24 hours striving to propose a solution and mediation plan within a week.

In any of the above channels, once the project proponent receives a report on the appeal, the project proponent should contact and discuss with the relevant community or other stakeholders within 24 hours, striving to propose a solution and mediation plan within one week. If the problem cannot be resolved within 30 days, it is recommended that stakeholders consider filing a lawsuit.

2.1.5 Public Comments

The public comment period of the project is from 19-November-2024 to 19-December-2024. Four comments were received during the public comment period.

<https://registry.verra.org/app/projectDetail/VCS/5313>

Comments received	Actions taken
1. [PG 50] Noting that the PD has opted for the Simple OM calculation and has chosen to employ the ex ante option for its operation margin calculation, i.e., the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the mission factor during the crediting period is required. a. It will be imperative for Verra team to request for a rationale for the decision to have opt for the ex-ante option, given that China's renewable energy mix in its grid is expected to increase overtime (at least, substantial change is expected within the project's 10-year crediting lifetime). b. It appears to me that ex-post option would provide a more	Firstly, in China, the data for OM calculations is not available. Therefore, we can only use the latest available OM data published by DNA (in China, the DNA is Ministry of Ecology and Environment). In the OM calculation file released by DNA on 08-July-2024, DNA clearly stated in accordance with the requirements of Tool to calculate the emission factor for an electricity system (version 07.0) that the OM emission factors for the Central China Power Grid in 2023 were calculated using the simple OM method, and were ex-ante calculated. Secondly, by combining the available OM emission factor data for CCPG over the past three

accurate accounting methodology of the GHG emissions that have been avoided through the displacement of the grid electricity.	years (2021⁸: 0.7938 tCO₂/MWh, 2022⁹: 0.8075 tCO₂/MWh, 2023: 0.8771 tCO₂/MWh), it was found that the values were gradually increasing. Based on this trend, the OM emission factor of CCPG is likely to increase after 2023. If applied ex-post option, the calculation of the reduction amount will also increase. Therefore, in order to conservative calculation, this project adopts ex-ante option. In conclusion, it is reasonable for this project to adopt the Simple OM and ex-ante calculation method, and it also meets the requirements of the methodology and the tools.
2. [PG 52] Noting that there is electricity exchange between Central China Power Grid and the Southern Power Grid, and that the emission factors are released nationally by the Chinese authorities. Based on my interpretation, the calculation of the adjusted grid emission factor used based on a weightage average of the annual power supply of the grid across all 7 power grids in China was considered, but not adopted – since the eventual OM grid emission factor used was 0.8771 tCO₂e/MWh, which is the OM grid emission factor for Central China (link). a. I would like to respectfully challenge whether a weightage average of Central China Power Grid and Southern Power Grid should be taken into consideration, in which the PD can consider using the lower of (average emission factors of Central China Grid 2023 – 2021 since Central China National Grid is the injection point of the project) and (weighted average emission factor across both Central China	Firstly, Ministry of Ecology and Environment of the People’s Republic of China has published a delineation of the project electricity system and connected electricity systems, so the project adopt the delineation of project electricity system and connected electricity system published by Ministry of Ecology and Environment of the People’s Republic of China. The power generated by the project displaces the equivalent electricity generated by Central China Power Grid. Secondly, Central China Power Grid is a larger regional grid, which consists of 4 sub-grids: Henan Province, Hubei Province, Hunan Province and Jiangxi Province. In addition, there is net imported power to Central China Power Grid from the North China Power Grid, Southwest China Power Grid and Northwest China Power Grid. According to the “Tool to calculate the emission factor for an electricity system”, the project chose option (b) to calculate the CO₂ emission factor for net electricity imports from the Central China Power Grid.

⁸ <https://ccer.cets.org.cn/notice/noticeDetail?bulletinInfold=1175122354980917248>

⁹ <https://ccer.cets.org.cn/notice/noticeDetail?bulletinInfold=1259912674523877376>

and Southern Power Grids in China 2023 – 2021). Rationale being, from a more conservative perspective, doing a lower of both methodologies will ensure that the emissions avoided is not overinflated.	Thirdly, according to data released by DNA, the OM factor of CCPG was the weighted average of the simple OM emission factors or adjusted simple OM emission factors for each of the last three years (2019-2021) for which data was available, with the power grid supply as the weight (included net imported power from the North China Power Grid, Southwest China Power Grid and Northwest China Power Grid). Similarly, the OM factor of Southern China Power Grid was also the weighted average of the simple OM emission factors or adjusted simple OM emission factors for each of the last three years (2019-2021) for which data was available, with the power grid supply as the weight (included net imported power from the Central China Power Grid and East China Power Grid). If the project considered a weightage average of Central China Power Grid and Southern Power Grid, the OM factor will be double-counted. Therefore, the project chose OM factor of CCPG to calculate is reasonable and in line with Tool 07.
3. [PG 77] Noting that for most other data/parameters, feasibility study report (FSR) is used as the basis of justification. However, the “Energy Conversion Efficiency of the project equipment” used the 40%, which is a default factor. I would like to clarify the rationale of the Project Developer of using so – given that the FSR should have utilised or consulted industry players on the efficiency range.	According to technical agreement of the generator set, the Energy Conversion Efficiency is 33%-36%. The project used 40% is more conservative. The reasons are as follows. According to AMS.III.G. (Version 10.0), the project can calculate $F_{CH4,PJ,y}$ using equation (6), in this equation, the Energy Conversion Efficiency (EE_y) appears as the denominator. If the project uses 33%-36% to calculate, the result is larger than using 40%. Therefore, the project uses 40% is more conservative. And it is also in line with the requirements of AMS.III.G. (Version 10.0).
4. Finally, in the entire PDD, there was no mention of treatment of transmission/distribution losses, which should result in 5 – 15% of reduction of the eventual consumption of total renewable electricity generated from the plant, hence	As mentioned in the methodology ACM0001 (version 19.0) applicable to large-scale project, when calculating $BE_{EC,y}$, the equation (2) in Tool 5, namely $BE_{EC,y} = EC_{BL,k,y} \times EF_{EF,k,y} \times (1 + TDL_{k,y})$, needs to be used. When calculating

it might be helpful for the PD to reach out to the provincial grid authority to have a estimative range of transmission losses, so that this can also be accounted for in the estimated tCO₂e avoided via energy source replacement.

PE_{EC,y}, the equation (1) in Tool 5 needs to be used, that is, $PE_{EC,y} = \sum_j EC_{PJ,j,y} * EF_{EL,j,y} * (1 + TDL_{j,y})$. However, this project is a small-scale project. Methodologies AMS-I.D: Grid connected renewable electricity generation (version 18.0) and AMS-III.G: Landfill methane recovery (version 10.0) are applied in this project. According to AMS-I.D (version 18.0), when calculating Baseline emissions include only CO₂ emissions from electricity generation in powerplants that are displaced due to the project activity, equation (1) is used, that is, $BE_y = EG_{PJ,y} \times EF_{grid,y}$. According to AMS-I.D (version 18.0) and AMS-III.G (version 10.0), the project emissions for this project are zero (the specific reasons have been described in Section 4.2 of this document and will not be repeated here). In summary, this project calculates the emission reduction based on the equations of AMS-I.D (version 18.0) and AMS-III.G (version 10.0). The equations do not involve the parameters of transmission and distribution losses. Therefore, the calculations for this project are reasonable and comply with the methodological requirements.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project proponent has successfully operated several recovery to power plants in China. It has also successfully developed carbon projects related to LFG recovery to power under VCS programme (VCS reference number: 2353, 2731, 2840). The employees of the project have work experience of community management more than 1 year, which is benefit for the project to communicate with relative stakeholders. Therefore, the PP has expertise and experience in implementing similar project activities.

2.2.2 Risk Assessment

Risks identified	Mitigation or preventative measure(s) taken
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Natural and human-induced risks to stakeholders' wellbeing	No risk identified	1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected; 2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 75,670 MWh, which could replace the equivalent electricity from fossil fuel based grid. The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. Thus, there are no natural and human-induced risks to stakeholders' wellbeing.
Risks to stakeholder participation	No risk identified	Based on the stakeholder meeting, the stakeholders involved in the project are very supportive of the project. The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China.
Working conditions	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. The Project has DCS system. Therefore, most of the employees work in indoor environment (at the office), instead of having to stand outside. In case of on-site monitoring and device maintenance, the project proponent provides safety protection equipment.
Safety of women and girls	No risk identified	The investment, construction and operation of the project are in compliance with the

		relevant laws and regulations of the People's Republic of China. China has published "Women's Rights Protection Law" ¹⁰ to provide support and protection to women and girls.
Safety of minority and marginalized groups, including children	No risk identified	The investment, construction and operation of the project are in compliance with the relevant laws and regulations of the People's Republic of China. China has published protection laws on the minors ¹¹ . Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not employ and is not complicit in any form of child labour. According to the Employee list, it could be confirmed that the Project does not employ any child labour.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	An environmental impact assessment (EIA) report of the project was approved by Enshi Tujia and Miao autonomous Prefecture Ecological Environment Bureau Lichuan City branch on 18-April-2024. During construction phase, the raise dust, noise, wastewater and solid wastes caused by project construction will be treated according to the measures in EIA report, and there will be no significant impact on the environment. During the operation period, all the mitigation measures proposed by EIA report will be implemented and the following key aspects will be addressed:

¹⁰ https://www.gov.cn/guoqing/2021-10/29/content_5647634.htm

¹¹ http://www.gov.cn/xinwen/2020-10/18/content_5552113.htm

Waste water

The wastewater of this project is mainly biogas condensate water and domestic sewage. Biogas condensate is collected by sewage tank, and domestic sewage is pretreated by septic tank. These waste water then enters the landfill leachate treatment station, after treatment to meet the “Domestic Waste Landfill Pollution Control Standard” (GB16889-2008) and then into the municipal sewage network.

Waste gas

The air pollutants produced by this project are mainly methane combustion waste gas and odor gas produced in the process of landfill gas collection and transport. After the combustion exhaust gas is treated to meet the standard, it is discharged through 15m exhaust cylinder. The emission concentration is in line with the emission limit of “Air Pollutant Emission Standard for Thermal Power Plant” (GB13223-2011). The odor gas is transported by closed pipeline, and the emission is small, which meets the secondary standard of “Odor Pollutant Emission Standard” (GB14554-93).

Noise pollution

Noise sources include equipment noise generated during the operation of drilling RIGS, air compressors, generator sets and various pumps. Through the selection of low noise equipment, reasonable layout, sound insulation, shock absorption and other measures, the factory boundary noise can meet the “Industrial Enterprise Factory Boundary Environmental Noise Emission Standard” (GB12348-2008) class 2 standards.

		Solid waste Solid waste includes domestic waste, general solid waste and hazardous waste. The particulate matter and waste filter element produced in the process of biogas purification belong to general solid waste. The particulate matter is collected and sent to the landfill for treatment, and the waste filter element is collected and returned to the manufacturer for recycling. Waste oil and waste catalyst are classified as hazardous waste. They are collected separately and temporarily stored in hazardous waste storage before being disposed of by qualified units. Domestic waste is collected in garbage bins and sent to landfill for disposal.
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2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ¹²	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	No discrimination was occurred. China has published “Labor Law” ¹³ and “Women’s Rights Protection Law”, which could prohibit such phenomena. The project abides the rules of equality and does not involve and is not complicit in any form of discrimination. According to the list of employees, both men and women are included, so this project does not involve discrimination.
Sexual harassment	No risk identified	No sexual harassment was occurred. China has published “Labor Law” and “Women’s Rights Protection Law”, which could prohibit such phenomena. The program adheres to

¹² The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

¹³ https://www.gov.cn/banshi/2005-05/25/content_905.htm

		the principle of respecting women, does not involve and is not complicit in any form of sexual harassment.
Equal pay for equal work	No risk identified	As a project combined of positive environmental, economic, and sustainable development benefits, the project is to generate clean electricity from the LFG sources to replace fossil fuel power. Qualified local residents, both men and women, are recruited to work for the project. The project follows the provisions of the “Constitution”¹⁴ and “Labor Law” on equal pay for men and women for equal work. During stakeholders’ consultation process, comments were collected from the local people, including both men and women.
Gender equity in labor and work	No risk identified	As a project combined of positive environmental, economic, and sustainable development benefits, the project is to generate clean electricity from the LFG sources to replace fossil fuel power. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in “Women’s Rights Protection Law” to provide support and benefits to women. Qualified local residents, both men and women, are recruited to work for the project. The project follows the provisions of the “Constitution”¹⁵ and “Labor Law” on equal pay for men and women for equal work. During stakeholders’ consultation process, comments were collected from the

¹⁴ https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm

¹⁵ https://www.gov.cn/guoqing/2018-03/22/content_5276318.htm

		local people, including both men and women. The project abides the rules of equality. Regarding the gender equality, detailed enforcement rules are regulated in “Labor Law” and “Women’s Rights Protection Law”.
Forced labor	No risk identified	There will be no forced labor in this project. China has promulgated the “Labor Law” to protect the legitimate rights and interests of workers and prohibit forced labor.
Child labor	No risk identified	China has published protection laws on the minors. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the power plant. Therefore, the project does not employ and is not complicit in any form of child labour. According to the Employee list, it could be confirmed that the Project does not employ any child labour.
Human trafficking	No risk identified	The project will not use victims of human trafficking. Human trafficking is a serious crime that is strictly regulated and combated in the Chinese legal system. The relevant laws and regulations mainly include the “Criminal Law”¹⁶ and the “Criminal Procedure Law”¹⁷.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	Human rights protection is formally embodied in the regulations in China. China fully recognizes international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples. China has its

¹⁶ <https://flk.npc.gov.cn/detail2.html?ZmY4MDgxODE3OTZhNjM2YTAxNzk4MjJhMTk2NDBjOTI>

¹⁷ <https://flk.npc.gov.cn/detail2.html?ZmY4MDgwODE2ZjEzNWY0NjAxNmYxZDFiODFiMDEzNTE%3D>

	own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties--the International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labor Law. This project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in “Women’s Rights Protection Law” to provide support and benefits to women. The project proponent and its activities are strictly regulated by local and national regulations, which obligate the project activities to in line with the international human right law and conventions. The legal approval by authorities demonstrate that the project does not and will not violate the regulations. According to the relevant provisions of the Labor Law and Women’s Rights Protection Law of China, the project continues to recognize, respect, and promote the protection of the rights of IPs, LCs, and customary rights holders.
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2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	According to the EIA report, no indigenous peoples present in or within the area of influence of the Project. According to the EIA report, the Project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
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No risk identified	The feasibility study report of the project was approved by the Development and Reform Bureau of Lichuan City on 27-February-2024. An environmental impact assessment (EIA) report of the project was approved by Enshi Tujia and Miao autonomous Prefecture Ecological Environment Bureau Lichuan City branch on 18-April-2024. The project is located at Lichuan landfill site, No.1 Group, Yangliu Community, Dongcheng Street Office, Lichuan City, Enshi Tujia and Miao Autonomous Prefecture, Hubei Province, P. R. China. According to the venue lease contract signed with Lichuan City Environmental Sanitation Service Center on 18-January-2024, the land used in this project is Lichuan landfill site, which does not involve land disputes. Based on questionnaire surveys and stakeholder consultation meetings, local residents have expressed support for the project. The project was constructed with the consent of government departments at all levels. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.
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2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not applicable as no property rights involved for the project.
Summary of the benefit sharing plan	Not applicable as no property rights involved for the project.
Approval and dissemination of benefit sharing plan	Not applicable as no property rights involved for the project.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The project is LFG recovery and utilization project. According to the EIA report, it does not affect biodiversity and ecosystems.

Soil degradation and soil erosion	No risk identified	The Project's purpose is to supply electricity from LFG. Therefore, the impact of the Project on soil degradation and soil erosion is negligible.
Water consumption and stress	No risk identified	The Project's purpose is to supply electricity from LFG. Therefore, the impact of the Project on water consumption and stress is negligible.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

Species and habitat	Not applicable as the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.
Areas needed for habitat connectivity	Not applicable as the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified	According to EIA, the project isn't adjacent to habitats for rare, threatened, or endangered species.
Areas for habitat connectivity	No risk identified	According to EIA, the project doesn't belong to areas for habitat connectivity.

2.4.2 Introduction of Species

According to EIA, no planting or species introduction are in the project. This section is left as N/A.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

According to EIA, no invasive species exist in the project area. Therefore, the following table isn't applicable.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	N/A

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No risk identified	According to EIA, no invasive species exist in the project area.

2.4.3 Ecosystem Conversion

The project isn't ARR, ALM, WRC or ACoGS project. Hence, not applicable.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	N/A

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.G.	AMS-III.G. Landfill methane recovery	10.0
Methodology	AMS-I.D.	AMS-I.D. Grid connected renewable electricity generation	18.0
Tool	04	Tool 04 Emissions from solid waste disposal sites	08.1
Tool	07	Tool 07 Tool to calculate the emission factor for an electricity system	07.0
Tool	08	Tool to determine the mass flow of a greenhouse gas in a gaseous stream	03.0
Tool	32	Tool 32 Positive lists of technologies	04.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
AMS-III.G.	2. This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.	Applicable. The Project consists in capturing landfill gas (which contains methane) from Lichuan landfill site, which is used for disposal of residues from human activities.
	3. Different options to utilise the recovered landfill gas as detailed in paragraph 4 for “AMS-III.H. : Methane recovery in wastewater treatment ” (version 19.0) are eligible for use under this methodology. The relevant procedures in AMS-III.H. shall be followed in this regard. Paragraph 4 of AMS-III.H.: The recovery biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed:	Applicable. The project utilizes the recovered LFG to generate electrical energy directly, i.e. 4(a) of AMS-III.H.(version 19.0).

	(i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	
	4. Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO₂ equivalent annually from all Type III components of the project activity.	Applicable. The project results in aggregate emission reduction of less than 60kt CO₂ equivalent annually from all Type III components.
	5. The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity	The implementation of the project does not reduce the amount of organic waste that would be recycled in the absence of the project. All the solid waste is disposed in Lichuan landfill site.
	6. This methodology is not applicable if the management of the solid waste disposal site (SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement). Such changes may include, for example, the addition of liquids to a SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic	Applicable. The management of the solid waste disposal site (SWDS) in the project activity is not deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity.

	degradation of the SWDS or changing the shape of the SWDS to increase methane production	
AMS-I.D.	4. This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant (s); (c) Involve a retrofit of (an) existing plant (s); (d) Involve a rehabilitation of (an) existing plant (s)/unit (s) or; (e) Involve a replacement of (an) existing plant(s).	Applicable. The project installs a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project, which corresponds to point (a).
	5. Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m²; (c) The project activity results in new reservoirs and the power density of the powerplant, as per definitions given in the project emissions section, is greater than 4W/m².	Not applicable. The project is not a hydro power plant.
	6. If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity	Not applicable. The project does not use non-renewable components nor co-fires fossil fuels. Anyway, the total

	applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	installed capacity is below 15 MW.
	7. Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable. The project only involves electricity generation.
	8. In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not applicable. The project does not involve addition of renewable energy generation units at an existing renewable power generation facility.
	9. In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.	Not applicable. The project does not involve retrofit, rehabilitation, or replacement.
	10. In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-L.C. Thermal energy production with or without electricity shall be explored.	Applicable. The project installed two engines to combust LFG of Lichuan landfill site, which mainly contains 50% methane. The baseline for the electricity component: equivalent electricity generated by the project is supplied by Central China Power Grid (CCPG), which is dominated by fossil fuel based power plants. The baseline for the electricity component is in accordance with procedure prescribed under 5.2 of methodology AMS-I.D.
	11. In case biomass is sourced from dedicated plantations, the applicability	Not applicable. The project does not involve biomass

	criteria in the tool “Project emissions from cultivation of biomass “shall apply.	sourced from dedicated plantations.
Tool 04	(a) Application A: The VCS project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "ACM0001: Flaring or use of landfill gas). The methane is generated from waste disposed in the past, including prior to the start of the VCS project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (VCS-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e. g. measuring the amount of methane captured from the SWDS).	Applicable. The project collects the methane from the Lichuan landfill site and uses it to generate the electricity.
	(b) Application B: The VCS project activity avoids or involves the disposal of waste at SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	Not applicable. The project activity mitigates methane emissions from Lichuan landfill site by capturing and utilizing the methane for power generation. This tool is only applied for ex ante estimation of the emissions in the PD.
Tool 07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity	Applicable. This tool is applied to calculate baseline emissions for power replace. Equivalent electricity generated by the project is

	to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects). Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity. In case of VCS projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country. Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	supplied by the CCPG, which is dominated by fossil fuel based power plants.
Tool 08	Typical applications of this tool are methodologies where the flow and composition of residual or flared gases or exhaust gases are measured for the	Applicable. This tool is applied to determination of baseline emissions by measuring the combusted LFG.

	determination of baseline or project emissions.	
	Methodologies where CO ₂ is the particular and only gas of interest should continue to adopt material balances as the means of flow determination and may not adopt this tool as material balances are the cost effective way of monitoring flow of CO ₂ .	Not applicable. The CH ₄ is the main gas in the landfill gas for the project.
	The underlying methodology should specify: (a)The gaseous stream the tool should be applied to; (b)For which greenhouse gases the mass flow should be determined; (c)In which time intervals the flow of the gaseous stream should be measured: and (d)Situations where the simplification offered for calculating the molecular mass of the gaseous stream (equations (3) or (17)) is not valid (such as the gaseous stream is predominantly composed of a gas other than N₂)	Applicable. The methodology has specified (a) to (d).
Tool 32	The use of this methodological tool is not mandatory for the project participants of a VCS project activity for demonstrating their additionality. This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool. The positive lists as contained in section 5 of this tool are valid up to 10 March 2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the “Procedure: Development, revision and clarification of	Applicable. The project collects the LFG from the Lichuan landfill site and uses it to generate the electricity, with a total capacity of 2 MW, which conforms to 5.1.1 (a) “The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW”.

baseline and monitoring methodologies and methodological tools”, there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date.

3.3 Project Boundary

According to the adopted methodology, the project boundary is determined as follows:

- The project boundary is the site of the project activity where the gas is captured and destroyed/used.
- If the electricity for project activity is sourced from grid or electricity generated by the LFG captured would have been generated by power generation sources connected to the grid, the project boundary shall include all the power generation sources connected to the grid to which the project activity is connected.
- If the electricity for project activity is from a captive generation source or electricity generated by the captured LFG would have been generated by a captive power plant, the captive power plant shall be included in the project boundary.

The project boundary is the site of the project activity, Lichuan landfill site, where the LFG is captured and used, and also includes all the power sources connected to the CCPG, which expands throughout Henan Province, Hubei Province, Hunan Province and Jiangxi Province.

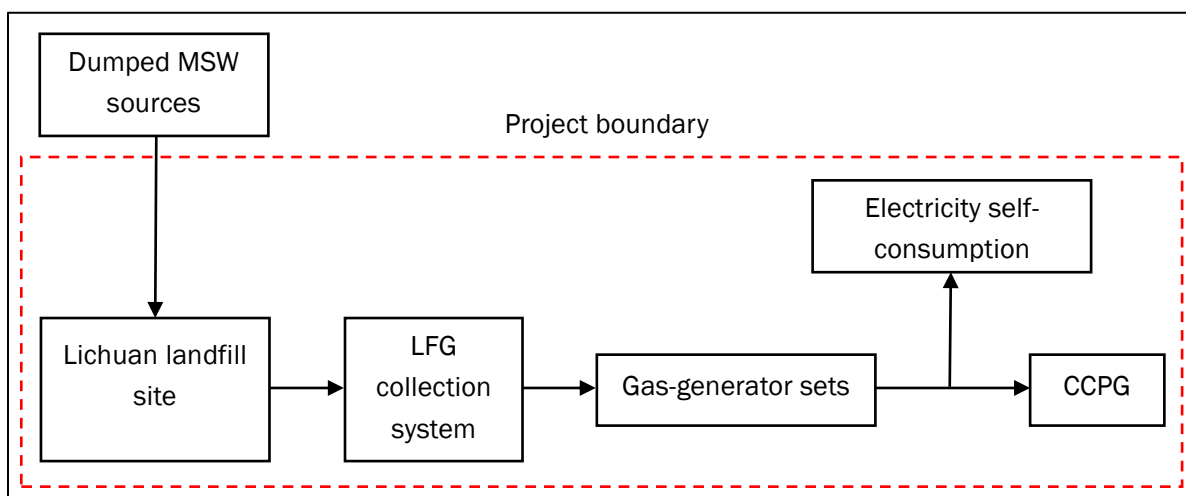


Figure 3.1 the Diagram of Project Boundary

Table 3.1 Summary of greenhouse gases and sources included in and excluded from the project boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from decomposition of waste at the SWDS site	CO ₂	No	CO ₂ emissions from decomposition of organic waste are not accounted since the CO ₂ is also released under the project activity
		CH ₄	Yes	The major source of emissions in the baseline
		N ₂ O	No	N ₂ O emissions are small compared to CH ₄ emissions from SWDS. This is conservative
	Emissions from electricity generation	CO ₂	Yes	Major emission source
		CH ₄	No	Excluded for simplification. This is conservative
		N ₂ O	No	Excluded for simplification. This is conservative
	Emissions from heat generation	CO ₂	No	No heat generation is included in the project activity
		CH ₄	No	No heat generation is included in the project activity
		N ₂ O	No	No heat generation is included in the project activity
	Emissions from the use of natural gas	CO ₂	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity
		CH ₄	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity

Source		Gas	Included?	Justification/Explanation
		N ₂ O	No	No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity
Project	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO ₂	No	Excluded for no fossil fuel is used by the project activity
		CH ₄	No	Excluded for no fossil fuel is used by the project activity
		N ₂ O	No	Excluded for no fossil fuel is used by the project activity
	Emissions from electricity consumption due to the project activity	CO ₂	Yes	May be an important emission source
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from flaring	CO ₂	No	Excluded for simplification
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
	Emissions from distribution of LFG using trucks and dedicated pipelines	CO ₂	No	No distribution of LFG using trucks and dedicated pipelines
		CH ₄	No	No distribution of LFG using trucks and dedicated pipelines
		N ₂ O	No	No distribution of LFG using trucks and dedicated pipelines

3.4 Baseline Scenario

According to the section 5.2. of the methodology AMS-III.G., the project chooses simplified procedures to identify the baseline scenario.

Landfill gas: In the absence of the project, MSW of Lichuan landfill site is left to decay within the project boundary, and methane is emitted to the atmosphere directly without any recovery and utilization.

Electricity: The project is a new grid-connected renewable power unit.

According to the section 5.3 of the methodology AMS-III.G. and 5.2 of methodology AMS-I.D., the baseline scenario is:

LFG from Lichuan landfill site is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by Central China Power Grid (CCPG), which is dominated by fossil fuel based power plants.

3.5 Additionality

The project uses the following steps to demonstrate additionality. The project meets the additionality criteria as outlined in the CDM methodology as:

Step 1: Regulatory Surplus: the project is not mandated by any legal requirements. Please refer to section 3.5.1 below for details.

Step 2: Positive List: the project conforms to the positive list as stipulated in the applies the simplified procedures in Section 5.2. of the methodology AMS-III.G. (version 10.0) and CDM “TOOL32: Positive lists of technologies” (version 04.0). Please refer to section 3.5.2 below for details.

The timeline of the project is as followed:

Time	Action
26-December-2023	Signed Project Cooperation Agreement
January-2024	Feasibility Study Report
27-February-2024	Project FSR Approval
20-March-2024	Board resolution
April-2024	EIA Report
18-April-2024	EIA Approval
19-April-2024	Stakeholder meeting
20-April-2024	Started construction
05-September-2024	Started operation

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

- ☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

- ☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

- ☐ Yes ☒ No

According to the relevant applicable laws and regulations at the time of submission of registration, i.e. Renewable Energy Law of the People's Republic of China which came into effect on 01-Jan-2006, Agenda for China's 14th Five-Year Plan (2021-2025), there are no new national and/or sectoral policies that could affect the baseline scenario during the crediting period. Therefore, the project is surplus to regulatory requirements.

3.5.2 Additionality Methods

According to section 5.2. of the methodology AMS-III.G. (version 10.0), project proponent may apply the following simplified procedure for additionality demonstration and baseline. The project chooses simplified procedures to demonstrate additionality. Tool 32 (Positive lists of technologies) is used to demonstrate additionality.

Requirements of tool 32 (Positive lists of technologies)	The project
5.1Waste handling and disposal 5.1.1Landfill gas recovery and its gainful use The project activities and PoAs at new or existing landfills (greenfield or brownfield)are deemed automatically additional if it is demonstrated that prior to the implementation of the project activities and PoAs the landfill gas (LFG) was only vented and or flared (in the case of brownfield projects)or would have been only vented and/or flared(in the case of greenfield projects)but not utilized for energy generation, and that under the project activities and PoAs any of the following conditions are met:	The LFG from Lichuan landfill site is vented to the atmosphere prior the implementation of the project for safety concerns. Total nameplate capacity of the project is 2MW, which is below 10MW.

(a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW; (b) The LFG is used to generate heat for internal or external consumption; (c) The LFG is flared.	
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Conclusion: the project is automatically additional.

3.6 Methodology Deviations

In this PD, the project used equation (22) to calculate $F_{CH_4,pj,y}$. This equation is not consistent with equation (6) in paragraph 21 of AMS-III.G. (version 10.0). The reasons are as follows. According to equation (4) of AMS-III.G. (version 10.0), when calculating ER, $F_{CH_4,pj,y}$ need to multiply “ GWP_{CH_4} ” here. If using equation (6) of AMS-III.G. (version 10.0) to calculate $F_{CH_4,pj,y}$, the finally calculated ER will multiply “ GWP_{CH_4} ” twice. The result calculated in this way is incorrect and was obtained through repeated calculations. $F_{CH_4,pj,y}$ means “Methane captured and destroyed/gainfully used by the project activity in the year y”, the unit is tCH₄. Equation (6) of AMS-III.G. (version 10.0) is themselves incorrect. The unit of the result calculated in this equation should be “tCO₂” not “tCH₄”. Therefore, in this PD, the error of this equation was corrected, i.e. remove “ GWP_{CH_4} ” from it, the project used equation (22) to calculate $F_{CH_4,pj,y}$.

The above deviation will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals.

The deviation is not involved in the criteria and procedures for monitoring or measurement and does not relate to any other part of the methodology.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The project utilizes the LFG for generation to substitute the equivalent electricity supplied by the grid, resulting in CH₄ and CO₂ emissions, which will be calculated as follows according to the methodology AMS-III.G. and AMS-I.D.

$$BE_{y,1} = \eta_{PJ} \times BE_{CH_4,SWDS,y} - (1 - OX) \times F_{CH_4,BL,y} \times GWP_{CH_4} \quad (1)$$

Where:

$BE_{y,1}$	=	Baseline emissions associated with the SWDS in year y (tCO ₂ e/yr)
η_{PJ}	=	Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50 per cent may be used
$BE_{CH_4,SWDS,y}$	=	Methane emission potential of a solid waste disposal site (in tCO ₂ e)
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering waste) (dimensionless). A default value of 0.1 may be used
$F_{CH_4,BL,y}$	=	Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (t CH ₄). The relevant procedures in “ACM0001: Flaring or use of landfill gas” may be followed, as well as taking into account the compliance with the relevant local laws and regulation if such laws and regulations exist
GWP_{CH_4}	=	Global Warming Potential for methane

Ex ante determination of $BE_{CH_4,SWDS,y}$

According to the AMS-III.G., $BE_{CH_4,SWDS,y}$ is calculated using the methodological tool “Emission from solid waste disposal sites”, which is calculated as follows:

$$BE_{CH_4,SWDS,y} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{k_j})) \quad (2)$$

Where:

$BE_{CH_4,SWDS,y}$	=	Baseline methane emissions occurring in the year y
φ_y	=	Model correction factor to account for model uncertainties for year y
f_y	=	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emission of methane to the atmosphere in year y
GWP_{CH_4}	=	Global Warming Potential of methane
OX	=	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
F	=	Fraction of the methane in the SWDS gas (volume fraction)

$DOC_{f,y}$	=	Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)
MCF_y	=	Methane correction factor for year y
$W_{j,x}$	=	Amount of organic waste type j disposed/prevented from disposal in the SWDS in the year y (t)
DOC_j	=	Fraction of degradable organic carbon in the waste type j (weight fraction for year y)
k	=	Decay rate for the waste type (1/yr)
j	=	Type of residual waste or types of waste in the MSW
x	=	Years of the crediting period for which waste is disposed at the SWDS, extending from the first year in the time period ($x=1$) to year y ($x=y$)
y	=	Years of the crediting period for which methane emissions are calculated

Determination of $F_{CH_4,BL,y}$

This section provides a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odour concerns, or for other reasons (collectively referred to as requirement in this section). The four cases in Table 4.1. are distinguished. The appropriate case should be identified, and the corresponding instructions followed.

Table 4.1. Cases for determining methane captured and destroyed in the baseline

Situation at the start of the project activity	Requirement to destroy methane	Existing LFG capture and destruction system
Case 1	No	No
Case 2	Yes	No
Case 3	No	Yes
Case 4	Yes	Yes

Currently China has regulations in place to deal with the management of landfills and to encourage utilization of LFG. Those regulations are:

“Standard for Pollution Control on the Landfill Site of Municipal Solid Waste” (GB 16889-2008), which became effective in 2008, issued by the Environment Protection Administration.

“Technical Code for Municipal Solid Waste Sanitary Landfill” (GB 50869-2013) , issued by the Ministry of Construction in 2013.

According to item 5.15 of GB16889-2008, if the designed landfill capacity is more than 2.5 million tons and the landfill thickness is more than 20m, methane utilization facilities or flare burning facilities shall be built to treat the landfill gas containing methane. For municipal solid waste

landfills smaller than the above scale, technologies that can effectively reduce methane generation and emission shall be adopted or flare combustion facilities shall be used to treat methane containing landfill gas.

Item 11.1.1 of GB 50869-2013 stipulates that the landfill site must be equipped with effective landfill gas drainage facilities to prevent the natural accumulation and migration of landfill gas, causing fire and explosion. Item 11.1.3 stipulates that if the landfill does not have the conditions for landfill gas utilization, the flare method shall be adopted for combustion treatment, and the process that can effectively reduce the generation and emission of methane shall be adopted. The old landfills that are not safe and stable should be equipped with effective landfill gas drainage facilities. Among them, item 11.1.1 is mandatory and must be strictly implemented.

In fact, the LFG of Lichuan landfill is emitted to atmosphere without LFG capture system prior the implementation of the project. Therefore, Case 2 listed in the table 3 is applicable for the project.

In this situation:

$$F_{CH_4,BL,y} = F_{CH_4,BL,R,y} \quad (3)$$

Where:

$F_{CH_4,BL,R,y}$ = Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (tCH₄/yr)

$F_{CH_4,BL,R,y}$ should be determined based on the information contained in the requirement to destroy methane, as follows:

- If the requirement specifies the amount of methane that must be flared then that amount is $F_{CH_4,BL,R,y}$;
- If the requirement specifies a percentage of the captured LFG that is required to be flared, the amount shall be calculated as follows:

$$F_{CH_4,BL,R,y} = \rho_{reg,y} \times F_{CH_4,PJ,capt,y} \quad (4)$$

Where:

$\rho_{reg,y}$ = Fraction of LFG that is required to be flared due to a requirement in year y

$F_{CH_4,PJ,capt,y}$ = Amount of methane in the LFG which is captured in the project activity in year y (tCH₄/yr)

The project proponent may choose to calculate $F_{CH_4,PJ,capt,y}$ by either of the two options:

- Option 1: Calculate using the “Tool to determine the mass flow of a greenhouse gas in a gaseous stream”, applying the following requirements:

- (i) The gaseous stream tool shall be applied to the LFG pipeline immediately downstream of the LFG capture system and before any split in the gaseous flow to different uses or flares;
 - (ii) CH₄ is the greenhouse gases for which the mass flow should be determined;
 - (iii) The simplification offered for calculating the molecular mass of the gaseous stream is valid; and
 - (iv) The mass flow should be calculated on an hourly basis for each hour h in year y .
- Option 2: Calculate as the sum of the amount of methane that is sent to the flare, electricity generating or heat generating equipment in year y as measured in section 5.4.1.1 in Methodology ACM0001, however, not taking into account the working hours of the equipment;
 - (i) If the requirement does not specify the amount or percentage of LFG that should be destroyed but requires the installation of a capture system, without requiring the captured LFG to be flared then:

$$F_{CH_4,BL,R,y} = 0 \quad (5)$$

- (ii) If the requirement does not specify any amount or percentage of LFG that should be destroyed, but requires the installation of a system to capture and flare the LFG, then a typical destruction rate of 20 per cent is assumed¹⁸:

$$F_{CH_4,BL,R,y} = 0.2 \times F_{CH_4,PJ,capt,y} \quad (6)$$

There is no requirement specifying the amount or percentage of LFG that should be destroyed, and no requirement on the installation of a system to capture and flare the LFG. As a result, it is assumed a typical destruction rate of 20 per cent by Option 2 (ii) to calculate the required LFG destruction.

Baseline emissions associated with electricity generation ($BE_{EC,y}$)

$$BE_{y,2} = EG_{PJ,y} \times EF_{grid,y} \quad (7)$$

Where:

$BE_{y,2}$	=	Baseline emissions associated with electricity generation in year y (t CO ₂)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh)

¹⁸ This default value of 20 per cent is based on assuming a situation in which: the efficiency of the LFG capture system in the project is 50 per cent; the efficiency of the LFG capture system in the baseline is 20 per cent; and, the amount captured in the baseline is flared using an open flare with a destruction efficiency of 50 per cent (consistent with the default value provided in the tool "Project emissions from flaring"). Project participants may propose and justify an alternative default value as a request for revision to this methodology.

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

Calculation of $EG_{PJ,y}$

The project is the installation of a greenfield power plant, therefore:

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad (8)$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y(MWh)

Calculation of $EF_{grid,y}$

According to the AMS-III.G., $EF_{grid,y}$ is calculated using the “Tool to calculate the emission factor for an electricity system” ($EF_{grid,y}=EF_{grid,CM,y}$).

Project participants shall apply the following six steps:

Step 1: Identify the relevant electricity systems;

Step 2: Choose whether to include off-grid power plants in the project

Step 3: Select a method to determine the operating margin (OM);

Step 4: Calculate the operating margin emission factor according to the selected method

Step 5: Calculate the build margin (BM) emission factor;

Step 6: Calculate the combined margin (CM) emission factor.

Step 1: Identity the relevant electricity systems

Ministry of Ecology and Environment of the People’s Republic of China has published a delineation of the project electricity system and connected electricity systems, so the project adopt the delineation of project electricity system and connected electricity system published by Ministry of Ecology and Environment of the People’s Republic of China. The power generated by the project displaces the equivalent electricity generated by Central China Power Grid. Central China Power Grid is a larger regional grid, which consists of 4 sub-grids: Henan Province, Hubei Province, Hunan Province and Jiangxi Province.

In addition, there is net imported power to Central China Power Grid from the North China Power Grid, Southwest China Power Grid and Northwest China Power Grid. According to the “Tool to calculate the emission factor for an electricity system”, use one of the following options to determine the CO₂ emission factor for net electricity imports from a connected electricity system:

- (a) OtCO_2/MWh ; or
- (b) The simple operating margin emission rate of the exporting grid, determined as described in Step 4, if the conditions for this method, as described in Step 3 below, apply to the exporting grid; or
- (c) The simple adjusted operating margin emission rate of the exporting grid, determined as described in Step 4 below; or
- (d) The weighted average operating margin (OM) emission rate of the exporting grid, determined as described in Step 4 below.

The PD will choose option (b) to calculate the CO_2 emission factor for net electricity imports from the Central China Power Grid.

According to the available data of 2020-2022¹⁹, the corresponding marginal emission factors of electricity are calculated, and weighted average is carried out to obtain the marginal emission factors of electricity.

Step2. Choose whether to include off-grid power plants in the project electricity system (optional).

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Following the calculation of the Ministry of Ecology and Environment of the People's Republic of China, and statistical data is available, Option I is chosen.

Step3. Select a method to determine the operating margin (OM)

"Tool to calculate the emission factor for an electricity system (Version 07.0)" offers four methods for the calculation of the operating margin emission factor(s) ($\text{EF}_{\text{grid,OM},y}$):

- (a) Simple OM; or
- (b) Simple adjusted OM; or
- (c) Dispatch data analysis OM; or
- (d) Average OM.

Method (a) -Simple OM is chosen for calculation and low-cost/must-run resources constitute less than 50% of the total grid generation in average of the five most recent years²⁰.

¹⁹ As of now, the data on grid baseline emission factors released by the Ministry of Ecology and Environment of China shows that the data for 2023 is the most recent and was released on 08-July-2024. In the released "Baseline Emission Factors for China's Regional Power Grid in 2023 Emission Reduction Projects" document, the data used for calculating OM originated from the period from 2020 to 2022.

²⁰ 2023 Bulletin on the Baseline Emission Factors of the China Grids

For simple OM, the emission factor can be calculated using either of the two following data vintages:

- (a) Ex ante option: If the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the mission factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the VCS-PD to the DOE for validation. For off-grid power plants, use a single calendar year within the 5 most recent calendar years prior to the time of submission of the VCS-PD for validation;
- (b) Ex post option: If the ex post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring. If the data required calculating the emission factor for year y is usually only available later than six months after the end of year y, alternatively the emission factor of the previous year y-1 may be used. If the data is usually only available 18 months after the end of year y, the emission factor of the year preceding the previous year y-2 may be used. The same data vintage (y, y-1 or y-2) should be used throughout all crediting periods.

Project participant employs ex ante option for its operation margin calculation.

Step4. Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated by one of the following two options:

- (a) Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit; or
- (b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if:
 - (i) The necessary data for Option A is not available; and
 - (ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and
 - (iii) Off-grid power plants are not included in the calculation (i.e., if Option I has been chosen in Step 2).

Since the data of each power plant/unit is unavailable, Option A is not applicable to the project. The project adopts Option B to calculate the operating margin emission factor ($EF_{grid,OM,y}$) of CCPG.

$$EF_{grid,OMsimple,y} = \frac{\sum_i (FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y})}{EG_y} \quad (9)$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO ₂ emission factor of fossil fuel type i in year y (tCO ₂ e/GJ), the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories for default values
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must run power plants/units in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

For Central China Power Grid, use the adjusted simple OM method to calculate their OM emission factor. The adjusted simple OM divides the generator sets into two categories: low cost/must run power plants and all the other power plants except low cost/ must run power plants. According to the weighted average of each unit's annual power supply emission factor, the probability weights of the two units participating in generation scheduling (λ and $1-\lambda$) are obtained. Finally, the unit power supply emission factor of the two types of generating units is adjusted according to the participation in generation. The probability weights of degrees are weighted and averaged to obtain the adjusted simple OM. The specific calculation formula is as follows:

$$EF_{grid,OM-adj,y} = (1 - \lambda_y) \times EF_{grid,OMsimple,y} + \lambda_y \times \frac{\sum_k (EG_{k,y} \times EF_{EL,k,y})}{\sum_k EG_{k,y}} \quad (10)$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh), according to the equation (6)
$EG_{k,y}$	=	Net electricity generated and delivered to the grid by low-cost/must run power plants/units type k in year y (MWh)

$EF_{EL,k,y}$	=	CO ₂ emission factor by low-cost/must run power plants/units type k in year y (tCO ₂ /MWh)
k	=	Type of low-cost/must run power plants/units in year y
λ_y	=	Probability weight of low-cost/must run power plants/units participating in generation scheduling in year y
y	=	The relevant year as per the data vintage chosen in Step 3

In the above formula, the unit power supply emission factor of the transferred electricity in the low-cost/must run unit adopts the transferred electricity OM emission factor of power grid. Hydropower, wind power, photovoltaic, geothermal power, low-cost biomass power, and nuclear power, the emission factor per unit power supply is 0 tCO₂/MWh. For low cost/must run units involved in power generation, the probability weight λ for scheduling is obtained by consulting “Tool to calculate the emission factor for an electricity system (Version 7.0)” Appendix 2 of Table 1.

Because there is also electricity exchange between Central China Power Grid and Southern Power Grid, the adjusted simple OM factor of the four power grids is obtained by solving a system of linear equations. In addition, since the OM factors of the seven regional power grids are calculated Ex ante, in accordance with the relevant provisions of the "Tool to calculate the emission factor for an electricity system" (version 07.0), the final OM emission factors of the regional power grids are reviewed and released by the competent authorities. Is the simple OM emission factor for each of the last three years (2019-2021) for which data is available or the value of the adjusted simple OM emission factor weighted by the weighted average of the annual power supply of the grid.

Regarding parameter selection, local values of $NCV_{i,y}$ and $EF_{CO_2,i,y}$ should be used where available. If no such values are available, IPCC default values are preferable. The Net Calorific Value ($NCV_{i,y}$) of each type of fossil fuel used in the calculation comes from China Energy Statistic Yearbook 2020. Emission factor ($EF_{CO_2,i,y}$) of each type of fossil fuel come from the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

The operating margin emission factor for 2019, 2020 and 2021 are calculated based on the data above. The three-year average is calculated as a weighted average of the emission factors. The Operational Margin Emission Factor is 0.8771 tCO₂/MWh.

Step5. Calculate the build margin emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

- (a) Option 1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of VCS PD submission to the VVB for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information

available on units already built at the time of submission of the request for renewal of the crediting period to the VVB. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

- (b) Option 2: For the first crediting period, the build margin emission factor shall be updated annually, ex-post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The project applies option 1 to calculate the build margin emission factor ex-ante.

According to the “Tool to calculate the emission factor for an electricity system”, the sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

- (a) Identify the set of five power units, excluding power units registered as VCS project activities, that started to supply electricity to the grid most recently ($SET_{5 \text{ units}}$) and determine their annual electricity generation ($AEG_{SET-5 \text{ units}}$, in MWh);
- (b) Determine the annual electricity generation of the project electricity system, excluding power units registered as VCS project activities (AEG_{total} , in MWh). Identify the set of power units, excluding power units registered as VCS project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET \geq 20\%}$, in MWh);
- (c) From $SET_{5 \text{ units}}$ and $SET_{\geq 20 \text{ percent}}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. Ignore steps (d), (e) and (f).

Otherwise:

- (d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as VCS project activity, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample-VCS}$) the annual electricity generation ($AEG_{SET-sample-VCS}$, in MWh);

If the annual electricity generation of that set is comprised at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-VCS} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample-VCS}$ to calculate the build margin. Ignore steps (e) and (f).

(a) Include in the sample group $SET_{sample-VCS}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(b) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sampleVCS} > 10yrs$)

The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m (EG_{m,y} \times EF_{EL,m,y})}{\sum_m EG_{m,y}} = \sum_m \left(\frac{EG_{m,y}}{\sum_m EG_{m,y}} \times EF_{EL,m,y} \right) \quad (11)$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

Under the current circumstances in China, the power plants consider the Build Margin data as important business data and will not have them published. Therefore, it is difficult to obtain the data of five power plants that have been put into operation most recently or the newly installed power plant capacity additions in the electricity system that comprise 20% of the system generation.

According to the instructions of China DNA, for the determination of the set of samples, a sample merging processing in some degree has been adopted due to that the power generation data, energy consumption data or thermal efficiency data of each plant cannot be consulted in the public statistical data. In this calculation, the newly-installed power units in the past years are classified by year, province and power generation technology, and the same type of newly installed power units in the same province and in the same year are bundled as a "newly-installed power units".

The power generation of each "newly-installed power units" in the most recent year y is estimated based on its installed capacity and the number of power generation utilization hours in year y . The formula is as follows:

$$EG_{m,y} = CAP_m \times H_{m,y} \quad (12)$$

Where:

$EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

CAP_m = Installed capacity of electricity generated and delivered to the grid by power unit m in year y (MW)

$H_{m,y}$ = The number of power utilization hours (h) of electricity generated and delivered to the grid by power unit m in year y . And it selects the average utilization hours of similar units in the province in which it is located in year y ;

y = Most recent year for which data is available

m = The sample group of power units

The power unit m is selected from the "newly-installed power plants" in the most recent year y (For the calculation of the grid BM in 2023, the y is equal to 2021) to the "newly-installed power plants" in the earlier year, until the cumulative power generation reaches 20% of the total power generation in the year y ($y=2021$).

Since the newly-installed power units of the same type (k) in the same province (A) and the same year (t) are bundled into the "newly-installed power units", the CAP_m is equal to the statistical data of recent installed capacity of a given unit type(k) in a given year(t) in a given province (A).

$$CAP_m = CAP_m|_{m=(A,t,k)} = CAP_{A,t,k} \quad (13)$$

Where:

CAP_m = Installed capacity of electricity generated and delivered to the grid by power unit m in year y (MW), and m is equivalent to an established combination of (A, t, k)

$CAP_{A,t,k}$ = Capacity of newly-installed power units of a given province (A), given year (t), and given unit type (k) (MW)

A = It is the various provincial regions covered by the regional power grid

t = It is the sampling year of the "newly-installed power units". For the calculation of the grid BM in 2023, t is equal to 2021, 2020, until the units that comprise at least 20 percent of the system generation in 2021.

k = It is the power generation technology classification of "newly-installed power units", which is divided into: hydro-power, coal-thermal power, gas-thermal power, oil-thermal power, Waste-thermal power plant, other thermal power²¹, nuclear power, wind power, solar power, and others²².

As per tool, the CO₂ emission factor of each power unit m ($EF_{EL,m,y}$) should be determined as per the tool in Step 4 section 6.4.1 for the simple OM, using Options A1, A2 or A3, using for y the most recent historical year for which electricity generation data is available, and using from the power units included in the build margin.

Because current statistics data cannot separate each power plant, for a power unit m , only data on electricity generation and the fuel types used is available. So, the option A2 is selected, the emission factor should be determined based on the CO₂ emission factor of the fuel type used and the efficiency of the power unit, as follows:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \quad (14)$$

Where:

- $EF_{EL,m,y}$ = CO₂ emission factor of power unit m in year y (tCO₂/MWh)
- $EF_{CO2,m,i,y}$ = Average CO₂ emission factor of fuel type i used in power unit m in year y (tCO₂/GJ)
- $\eta_{m,y}$ = Average net energy conversion efficiency of power unit m in year y (dimensionless)
- 3.6 = Conversion coefficient of thermal work equivalent of electricity (GJ/MWh)

According to Equation (11), the unit electricity emission factor of the hydro-power, nuclear power, wind power, solar power, other thermal power, and others power generation technology in the "newly-installed power units" samples are is zero. The emission factor per unit of electricity for power generation from coal, gas, oil and electricity waste power is calculated based on Equation (11). Since the average net energy conversion efficiency of each sample ($\eta_{m,y}$) cannot be obtained, the power supply thermal efficiency of the best commercialized technology of coal, gas, oil and waste power ($\eta_{Best,m,y}$) is better than $\eta_{m,y}$. It is conservative to use $\eta_{Best,m,y}$ for the calculation of $EF_{EL,m,y}$. Therefore, the formula is as follows:

$$EF_{Best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{Best,m,y}} \quad (15)$$

²¹ refers to waste heat and pressure, straw, bagasse, forest wood power generation

²² refers to power generation such as geothermal energy and ocean energy.

Where:

$EF_{Best,m,y}$ = Emission factor of power unit m with the best technology commercially available in year y (tCO₂/MWh)

$\eta_{Best,m,y}$ = Power generation efficiency of the best technology commercially available in year y (dimensionless)

m = Sample of "newly-installed power units" generated by coal, gas, oil and waste incineration in a province in a certain year

The emission factors of coal, natural gas, fuel oil and municipal solid waste (MSW) are sourced from the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, and the lower values of the 95% confidence interval are applied to be conservative.

No fuel oil power units were constructed in 2021. The power generation efficiency of the best technology commercially available for fuel oil units comes from Baseline Emission Factors for Regional Grids in China of previous years, i.e., 52.9%.

According to the above steps, the build margin emission factor, $EF_{grid,BM,y}$ of the CCPG is calculated to be 0.2696 tCO₂/MWh, which is confirmed by 2023 Baseline Emission Factors for Regional Power Grids in China published by Ministry of Ecology and Environment of the People's Republic of China²³.

Step6. Calculate the combined margin emission factor

Combined Margin emission factor ($EF_{grid,CM,y}$) is calculated as the weighted average of the operating margin emission factor ($EF_{grid,OM,y}$) and the build margin emission factor ($EF_{grid,BM,y}$), where the weights ω_{OM} and ω_{BM} , by default, are 0.5 and 0.5 in the first crediting period, and $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ are calculated as described above and are expressed in tCO₂/MWh.

$$EF_{grid,CM,y} = \omega_{OM} \times EF_{grid,OM,y} + \omega_{BM} \times EF_{grid,BM,y} \quad (16)$$

$$EF_{grid,CM,y} = 0.5 \times 0.8771 + 0.5 \times 0.2696 = 0.57335 \text{ (tCO}_2\text{/MWh)}$$

The $EF_{OM,y}$, $EF_{BM,y}$ and $EF_{CM,y}$ are ex-ante calculation and are fixed during the credit period.

By combining all the above equations, the ex-ante estimate of the emission reduction is:

$$BE_y = \eta_{PJ} \times BE_{CH_4,SWDS,y} - (1 - OX) \times F_{CH_4,BL,y} \times GWP_{CH_4} + EG_{PJ,y} \times EF_{grid,y} \quad (17)$$

4.2 Project Emissions

$$PE_y = PE_{power,y} + PE_{flare,y} + PE_{process,y} \quad (18)$$

Where:

²³ <https://ccer.cets.org.cn/notice/noticeDetail?bulletinInfold=1259913708889575424>

PE_y	=	Project emission in year y (t CO _{2e})
$PE_{power,y}$	=	Emission from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO _{2e})
$PE_{flare,y}$	=	Emissions from flaring or combustion of the landfill gas stream in the year y (t CO _{2e})
$PE_{process,y}$	=	Emission from the landfill gas upgrading process in the year y (t CO _{2e}), determined by following the relevant procedures described in annex 1 of AMS-III.H.

According to AMS-III.G., project emissions from electricity consumption are determined as per the procedures described in AMS-I.D. “Grid connected renewable electricity generation”. According to AMS-I.D., the project is neither involved geothermal power plants nor hydro power plants, project emissions from electricity consumption are identified as 0.

The project does not involve the emissions from the consumption of fossil fuels, only emissions from the use of electricity for the operation of the installed facilities are involved. The project will consume electricity delivered from the grid when the project is not in operation. And this part of electricity has already been deducted from the quantity of electricity supplied by the project plant/unit to the grid when calculating the Quantity of net electricity generation supplied by the project plant/unit to the grid, which is then used to calculate the Baseline emissions associated with electricity generation ($BE_{EC,y}$) hence emissions from the use of electricity for the operation of the installed facilities are excluded, which means Emissions from the use of fossil fuel or electricity for the operation of the installed facilities equal to 0. Therefore, $PE_{power,y}$ is equal to 0.

There is no flare used to destroy the LFG, therefore, $PE_{flare,y}$ is equal to 0.

The project captures LFG for power generation and not involved upgrading process, therefore, $PE_{process,y}$ is equal to 0.

In conclusion, the project emission of the project activity is 0.

4.3 Leakage Emissions

There is no equipment transferred in the project, no leakage effects need to be accounted under AMS-III.G. and AMS-I.D.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

According to AMS-III.G., the emission reduction achieved by the project activity can be estimated ex-ante in the PD by:

$$ER_{y,estimated} = BE_y - PE_y - LE_y \quad (19)$$

By combining all the above equations, the ex-ante estimate of the emission reduction is:

$$ER_{y,estimated} = \eta_{PJ} \times BE_{CH_4,SWDS,y} - (1 - OX) \times F_{CH_4,BL,y} \times GWP_{CH_4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad (20)$$

Emission reductions (ex-post)

According to AMS-III.G., the actual emission reduction achieved by the project activity during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the project activity, calculated as:

$$ER_{y,calculated} = (1 - OX) \times (F_{CH_4,PJ,y} - F_{CH_4,BL,y}) \times GWP_{CH_4} - PE_y - LE_y \quad (21)$$

Where:

$F_{CH_4,PJ,y}$ = Methane captured and destroyed/gainfully used by the project activity in the year y (t CH₄)

The project utilizes the recovered methane for power generation. Therefore $F_{CH_4,PJ,y}$ is calculated as follows, based on the amount of monitored electricity generation without monitoring methane flow and concentration:

$$F_{CH_4,PJ,y} = \frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4} \quad (22)^{24}$$

Where:

EG_y = Electricity generation in year y (MWh)

3600 = Conversion factor (1 MWh=3600 MJ)

NCV_{CH_4} = NCV of methane (MJ/NM³) use default value: 35.9 MJ/NM³

EE_y = Energy Conversion Efficiency of the project equipment determined from one of the following options:

- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel if the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
- Default efficiency of 40 per cent

Therefore:

²⁴ This equation is different with the equation (6) of the methodology AMS-III.G (version 10), because the calculation of $F_{CH_4,PJ,y}$ in the methodology AMS-III.G. (version 10.0) multiplies "GWP_{CH₄}" again, which is repeated with the calculation of $ER_{y,calculated}$ in equation (4) of AMS-III.G. (version 10.0). In order to calculate the conservatism of the results, the project used equation (22) to calculate $F_{CH_4,PJ,y}$.

$$ER_{y,calculated} = (1 - OX) \times \left(\frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4} - F_{CH_4,BL,y} \right) \times GWP_{CH_4} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad (23)$$

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
05-Sep-2024 to 31-Dec-2024	17,847	0	0	17,847	0	17,847
01-Jan-2025 to 31-Dec-2025	59,323	0	0	59,323	0	59,323
01-Jan-2026 to 31-Dec-2026	63,430	0	0	63,430	0	63,430
01-Jan-2027 to 31-Dec-2027	56,543	0	0	56,543	0	56,543
01-Jan-2028 to 31-Dec-2028	50,640	0	0	50,640	0	50,640
01-Jan-2029 to 31-Dec-2029	45,566	0	0	45,566	0	45,566
01-Jan-2030 to 31-Dec-2030	41,188	0	0	41,188	0	41,188
01-Jan-2031 to 31-Dec-2031	37,398	0	0	37,398	0	37,398
01-Jan-2032 to 31-Dec-2032	34,104	0	0	34,104	0	34,104
01-Jan-2033 to 31-Jan-2033	31,229	0	0	31,229	0	31,229
01-Jan-2024 to 04-Sep-2024	19,428	0	0	19,428	0	19,428
Total	456,696	0	0	456,696	0	456,696

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	OX
Data unit	-
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.1
Justification of choice of data or description of measurement methods and procedures applied	As the landfill was covered by soil, the default value for oxidation could be applied.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ e/tCH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC Fifth Assessment Report (AR5)
Value applied:	Default value of 28. Shall be updated according to any future COP/CMP decisions
Justification of choice of data or description of measurement methods and procedures applied	-

Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	ρ_{CH_4}
Data unit	t/m ³
Description	Density of methane gas at Normal Conditions
Source of data	Tool 08 “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0)
Value applied:	0.0007156 at normal conditions (273.15k and 101.325kPa)
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per equation (6) of the Tool 08 “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0). $\rho_{CH_4} = 101,325 \text{ Pa} \times 16.04 \text{ kg/mol} / 8,314 \text{ Pa.m}^3/\text{kmol.k} / 273.15 \text{ k}$ $= 0.7156 \text{ kg/m}^3$ $= 0.0007156 \text{ t/m}^3$
Purpose of data	Calculation of baseline emissions
Comments	At the normal conditions (273.15 K and 101,325 Pa)

Data / Parameter	η_{PJ}
Data unit	-
Description	Efficiency of the LFG capture system that will be installed in the project activity
Source of data	FSR of the project
Value applied:	85%

Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of baseline emissions
Comments	The data in the feasibility study report are calculated according to the area of the landfill area and the area of drilling Wells. It is based on a landfill gas calculation model published by the EPA ²⁵ . According to China's landfill gas estimation model ²⁶ , the collection rate is also 85%. The literature suggests that a garbage collection rate of 60-90% is reasonable. The landfill gas collection system can capture 60 to 90% of the produced CH ₄ (Dingman, 2009; US EPA, 2017b) ²⁷ . Therefore, it is reasonable to use FSR as a source of this data.

Data / Parameter	Φ_y
Data unit	-
Description	The model correction factor to account for model uncertainties
Source of data	Default value of the tool "Emissions from solid waste disposal sites" (version 08.1)
Value applied:	0.75
Justification of choice of data or description of measurement methods and procedures applied	Application A is used to decide the value. And the value of wet conditions for used for the project since the MAP/PET > 1. The meteorological data for the project site (Lichuan City) are: Mean annual temperature (MAT): 13.2 °C ²⁸ Mean annual precipitation (MAP): 1,400 mm ²⁹

²⁵ <https://www.epa.gov/lmop>

²⁶ https://globalmethane.org/documents/models/pdfs/UsersManualChinaLFGmodel_v1.1-chi.pdf

²⁷ D.M. Dingman-Applications and Potentials for Biogenic Methane Recovery Operations in Nebraska Agriculture, Industry, and Economic Development. Environmental Studies Undergraduate Theses (2009)
<https://digitalcommons.unl.edu/envstudtheses>

²⁸ http://www.lichuan.gov.cn/zjlc/zrzy/201908/t20190801_530866.shtml

²⁹ http://slj.enshi.gov.cn/ssjj/202308/t20230825_1475060.shtml

	Potential evapotranspiration (PET): 1,079.4 mm ³⁰
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	f_y
Data unit	-
Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
Source of data	The tool “Emissions from solid waste disposal sites” (version 08.1)
Value applied:	0
Justification of choice of data or description of measurement methods and procedures applied	With the factor “$f=0.0$” because the amount of LFG that would have been captured and destroyed is already accounted for in equation (1). For the project, the amount of LFG that would have been captured and destroyed is already counted for in the parameter $F_{CH_4,BL,y}$ in the equation. Therefore, $f_y=0$
Purpose of data	Calculation of baseline emissions
Comments	For Application A: f_y is calculated once for the crediting period.

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

³⁰ https://r.estv.com.cn/dtxresources/global/jcms/jcms_files/jcms1/web1/site/attach/0/f100aff2423f4967a64bb940887a0fc5.pdf

Value applied:	0.5
Justification of choice of data or description of measurement methods and procedures applied	This coefficient reflects the fact that some degradable organic wastes are not degraded or degrade very slowly in the anaerobic environment of landfill site. The default IPCC value is 0.5.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$DOC_{f,y}$
Data unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposed in the SWDS
Source of data	the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.5
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	MCF_y
Data unit	-
Description	Methane correction factor
Source of data	the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

Value applied:	1.0
Justification of choice of data or description of measurement methods and procedures applied	The project using application A to decide the value of MCF_y. In case of Application A, $MCF_y = 1.0$ for anaerobic managed solid waste disposal sites. These must have controlled placement of waste (i.e. waste directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires) and will include at least one of the following: (i) cover material; (ii) mechanical compacting; or (iii) levelling of the waste. Lichuan landfill site strictly complies with the relevant national laws and regulations to manage the landfill site in a standardized manner. The garbage is transported to the treatment site for separate landfilling. It undergoes processes such as hazardous waste monitoring, weighing, spreading and compaction, chemical disinfection, landfilling, covering, gas diversion and leachate treatment³¹. Therefore, MCF_y value of 1.0 for anaerobic managed solid waste disposal sites should be applied in line with “Emissions from solid waste disposal sites” (version 08.1)
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	$W_{j,x}$										
Data unit	t										
Description	Amount of solid waste type j disposed or prevent from disposal in the SWDS in year x										
Source of data	Feasibility study report (FSR) of the project										
Value applied:	<table border="1"> <thead> <tr> <th>Year</th><th>Annual landfilled MSW (tons)</th></tr> </thead> <tbody> <tr> <td>2007</td><td>5,873</td></tr> <tr> <td>2008</td><td>72,916</td></tr> <tr> <td>2009</td><td>74,404</td></tr> <tr> <td>2010</td><td>75,923</td></tr> </tbody> </table>	Year	Annual landfilled MSW (tons)	2007	5,873	2008	72,916	2009	74,404	2010	75,923
Year	Annual landfilled MSW (tons)										
2007	5,873										
2008	72,916										
2009	74,404										
2010	75,923										

³¹ Based on the statement of Lichuan landfill site.

	2011	76,690	
	2012	77,464	
	2013	78,247	
	2014	79,037	
	2015	79,836	
	2016	80,642	
	2017	81,457	
	2018	82,279	
	2019	83,111	
	2020	83,950	
	2021	125,208	
	2022	126,473	
	2023	127,750	
	2024	134,776	
	2025	142,189	
	2026	150,009	
	Component	Weight fraction (wet waste)	Remark
	W1-Wood and wood products	12.95%	Difficult-to-degrade
	W2-Pulp, paper and cardboard	12.9%	Degradable
	W3-Food, food waste, beverages and tobacco	37.36%	Easy-to-degrade
	W4-Textiles	1.57%	Difficult-to-degrade
	W5-Garden, yard and park waste	1.00%	Easy-to-degrade

	W6-Glass, plastic, metal other inert	34.22%	Non-degradable
Justification of choice of data or description of measurement methods and procedures applied	It is calculated as total waste amount dumped in the landfill site in the year x multiplied by organic waste type j fraction on wet basis. Both total waste amount and waste type j fraction are referred to FSR. The data was double checked with that given by Lichuan landfill site.		
Purpose of data	Calculation of baseline emissions		
Comments	-		

Data / Parameter	DOC _j															
Data unit	-															
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)															
Source of data	“Emissions from solid waste disposal sites” (version 08.1)															
Value applied:	<table><tr><td>Waste type j</td><td>DOC_j (% wet waste)</td></tr><tr><td>Wood and wood products</td><td>43</td></tr><tr><td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr><tr><td>Food, food waste, beverages and tobacco (other than sludge)</td><td>15</td></tr><tr><td>Textiles</td><td>24</td></tr><tr><td>Garden, yard and park waste</td><td>20</td></tr><tr><td>Glass, plastic, metal, other inert waste</td><td>0</td></tr></table>		Waste type j	DOC _j (% wet waste)	Wood and wood products	43	Pulp, paper and cardboard (other than sludge)	40	Food, food waste, beverages and tobacco (other than sludge)	15	Textiles	24	Garden, yard and park waste	20	Glass, plastic, metal, other inert waste	0
Waste type j	DOC _j (% wet waste)															
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Pulp, paper and cardboard (other than sludge)	40															
Food, food waste, beverages and tobacco (other than sludge)	15															
Textiles	24															
Garden, yard and park waste	20															
Glass, plastic, metal, other inert waste	0															

Justification of choice of data or description of measurement methods and procedures applied	Data available are based on the wet condition of the waste.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	k_j		
Data unit	1/yr		
Description	Decay rate for the waste type j		
Source of data	"Emissions from solid waste disposal sites" (version 08.1)		
Value applied:	Waste type j		k_j (MAT ≤ 20 °C, MAP/PET > 1)
	Slow degrading	Pulp, paper and cardboard (other than sludge), textiles	0.06
		Wood, wood products and straw	0.03
	Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.10
	Rapidly degrading	Food, food waste, sewage sludge, beverages and tobacco	0.185
Justification of choice of data or description of measurement methods and procedures applied	The meteorological data for the project site (Lichuan City) are: Mean annual temperature (MAT): 13.2 °C³² Mean annual precipitation (MAP): 1,400 mm³³ Potential evapotranspiration (PET): 1,079.4 mm³⁴		

³² http://www.lichuan.gov.cn/zjlc/zrzy/201908/t20190801_530866.shtml

³³ http://slj.enshi.gov.cn/ssjj/202308/t20230825_1475060.shtml

³⁴ https://r.estv.com.cn/dtxresources/global/jcms/jcms_files/jcms1/web1/site/attach/0/f100aff2423f4967a64bb940887a0fc5.pdf

Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{grid,OM,y}
Data unit	tCO ₂ /MWh
Description	Operation margin emission factor of CCPG
Source of data	2023 China baseline emission factor of regional power grid
Value applied:	0.8771
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Energy Statistic Yearbook (2020-2022), System of Statistical Survey of Energy Consumption in Public Institutions (National Government Offices Administration, approved by the National Bureau of Statistics, August 2019), China Electric Power Yearbook (2020-2022), Power Industry Statistical Data Collection (2019-2021) and IPCC data.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{grid,BM,y}
Data unit	tCO ₂ /MWh
Description	Build margin emission factor of CCPG
Source of data	2023 China baseline emission factor of regional power grid
Value applied:	0.2696
Justification of choice of data or description of	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Energy Statistic Yearbook (2020-2022), System of Statistical Survey of Energy

measurement methods and procedures applied	Consumption in Public Institutions (National Government Offices Administration, approved by the National Bureau of Statistics, August 2019), China Electric Power Yearbook (2020-2022), Power Industry Statistical Data Collection (2019-2021) and IPCC data.
Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	NCV _{CH4}
Data unit	MJ/NM ³
Description	Net calorific value of methane at reference conditions
Source of data	AMS-III.G. "Landfill methane recovery" (version 10.0).
Value applied:	35.9 MJ/NM ³
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of data	Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	F _{CH4,BL,y}
Data unit	t CH ₄ /yr
Description	Amount of methane in the LFG which is flared due to a requirement in year y
Source of data	Information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odor concerns.

Description of measurement methods and procedures applied	-																								
Frequency of monitoring/recording	Annually																								
Value applied:	The data below are calculated ex ante: <table border="1"> <thead> <tr> <th>Year</th><th>Amount of methane in the LFG which is flared due to a requirement (tCH₄)</th></tr> </thead> <tbody> <tr> <td>05-Sep-2024 to 31-Dec-2024 (both days included)</td><td>141</td></tr> <tr> <td>2025</td><td>468</td></tr> <tr> <td>2026</td><td>500</td></tr> <tr> <td>2027</td><td>446</td></tr> <tr> <td>2028</td><td>399</td></tr> <tr> <td>2029</td><td>359</td></tr> <tr> <td>2030</td><td>325</td></tr> <tr> <td>2031</td><td>295</td></tr> <tr> <td>2032</td><td>269</td></tr> <tr> <td>2033</td><td>246</td></tr> <tr> <td>01-Jan-2034 to 04-Sep-2034 (both days included)</td><td>153</td></tr> </tbody> </table>	Year	Amount of methane in the LFG which is flared due to a requirement (tCH ₄)	05-Sep-2024 to 31-Dec-2024 (both days included)	141	2025	468	2026	500	2027	446	2028	399	2029	359	2030	325	2031	295	2032	269	2033	246	01-Jan-2034 to 04-Sep-2034 (both days included)	153
Year	Amount of methane in the LFG which is flared due to a requirement (tCH ₄)																								
05-Sep-2024 to 31-Dec-2024 (both days included)	141																								
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2031	295																								
2032	269																								
2033	246																								
01-Jan-2034 to 04-Sep-2034 (both days included)	153																								
Monitoring equipment	-																								
QA/QC procedures applied	The calibration should be done once a year by a qualified third party.																								
Purpose of data	Calculation of baseline emissions																								
Calculation method	The data is calculated as methane capture and destroyed multiplied by fraction of LFG that is required to be flared due to a requirement. Methane destroyed is calculated based on electricity generation.																								

Comments	Applicable to Case 2 of section 5.4.1.3 of ACM0001 “Flaring or use of landfill gas” (version 19.0)
Data / Parameter	$\rho_{reg,y}$
Data unit	Dimensionless
Description	Fraction of LFG that is required to be flared due to a requirement in year y
Source of data	Information of the host country’s regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odor concerns.
Description of measurement methods and procedures applied	According to the methodology ACM0001 (version 19.0), if the requirement does not specify any amount or percentage of LFG that should be destroyed, but requires the installation of a system to capture and flare of the LFG, then a typical destruction rate of 20 percent is assumed.
Frequency of monitoring/recording	Annually
Value applied:	20%
Monitoring equipment	-
QA/QC procedures applied	Updated annually according to information of the host country’s regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odor concerns.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	Applicable to Case 2 of section 5.4.1.3 of ACM0001 “Flaring or use of landfill gas” (version 19.0)
Data / Parameter	$EG_{PJ, facility,y}$

Data unit	MWh																										
Description	Quantity of net electricity generated supplied by the project plant/unit in year y																										
Source of data	Electricity meter E3																										
Description of measurement methods and procedures applied	Measured continuously by a bi-directional electricity meter installed at the project site. All data will be monitored and archived electronically. Double check by receipt of electricity sales.																										
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.																										
Value applied:	The data below are calculated ex ante <table><tr><th>Year</th><th>Quantity of net electricity generated supplied by the project plant (MWh)</th></tr><tr><td>05-Sep-2024 to 31-Dec-2024 (both days included)</td><td>2,957</td></tr><tr><td>2025</td><td>9,829</td></tr><tr><td>2026</td><td>10,510</td></tr><tr><td>2027</td><td>9,368</td></tr><tr><td>2028</td><td>8,391</td></tr><tr><td>2029</td><td>7,550</td></tr><tr><td>2030</td><td>6,825</td></tr><tr><td>2031</td><td>6,197</td></tr><tr><td>2032</td><td>5,651</td></tr><tr><td>2033</td><td>5,174</td></tr><tr><td>01-Jan-2034 to 04-Sep-2034 (both days included)</td><td>3,219</td></tr></table>			Year	Quantity of net electricity generated supplied by the project plant (MWh)	05-Sep-2024 to 31-Dec-2024 (both days included)	2,957	2025	9,829	2026	10,510	2027	9,368	2028	8,391	2029	7,550	2030	6,825	2031	6,197	2032	5,651	2033	5,174	01-Jan-2034 to 04-Sep-2034 (both days included)	3,219
Year	Quantity of net electricity generated supplied by the project plant (MWh)																										
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2031	6,197																										
2032	5,651																										
2033	5,174																										
01-Jan-2034 to 04-Sep-2034 (both days included)	3,219																										
Monitoring equipment	Electricity meter E3																										
	Type	Accuracy class	Serial number																								

	DSZ331	0.5S	10007000875692
QA/QC procedures applied	The calibration should be done once a year by a qualified third party.		
Purpose of data	Calculation of baseline emissions		
Calculation method	-		
Comments	-		

Data / Parameter	EG _y	
Data unit	MWh	
Description	Electricity generation in year y	
Source of data	Electricity meters E1 and E2	
Description of measurement methods and procedures applied	Measured continuously by electricity meter installed at the project site. All data will be monitored and archived electronically.	
Frequency of monitoring/recording	The recording frequency will be hourly measured and record, and monthly aggregated.	
Value applied:	The data below are calculated ex ante	
	Year	Electricity generated in year y (MWh)
	05-Sep-2024 to 31-Dec-2024 (both days included)	3,146
	2025	10,457
	2026	11,180
	2027	9,966
	2028	8,926
	2029	8,032

	2030	7,260
	2031	6,592
	2032	6,011
	2033	5,505
	01-Jan-2034 to 04-Sep-2034 (both days included)	3,425
Monitoring equipment	Electricity meter E1	
	Type	Accuracy class
	DSZ747	0.5S
	Serial number	
	00100006356956	
	Electricity meter E2	
	Type	Accuracy class
	DSZ747	0.5S
	Serial number	
	00100006356952	
QA/QC procedures applied	The calibration should be done once a year by a qualified third party.	
Purpose of data	Calculation of baseline emissions	
Calculation method	-	
Comments	-	

Data / Parameter	D _{CH₄,y}
Data unit	t/m ³
Description	Density of methane at the temperature and pressure of the landfill gas in year y (tonnes/m ³). If LFG _{i,y} is reported at normal conditions of temperature and pressure, the density of methane is also determined at normal conditions
Source of data	Tool 08 “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0)

Description of measurement methods and procedures applied	The project employs landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas is displayed. According to the AMS-III.G., the density of methane is also determined at normal conditions
Frequency of monitoring/recording	-
Value applied:	0.0007156
Monitoring equipment	-
QA/QC procedures applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	P
Data unit	Pa
Description	Pressure of the landfill gas
Source of data	-
Description of measurement methods and procedures applied	The pressure of the gas is required to determine the density of the methane combusted. The project employs landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas is displayed.
Frequency of monitoring/recording	-
Value applied:	-

Monitoring equipment	-
QA/QC procedures applied	Operators must receive training to master skills in equipment installation, maintenance, and troubleshooting, avoiding deviations caused by human operational errors.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	If the landfill gas flow meter employed measures flow, pressure and temperature and displays or outputs the normalized flow of landfill gas, then there is no need for separate monitoring of pressure and temperature of the landfill gas.

Data / Parameter	T
Data unit	°C
Description	Temperature of the landfill gas
Source of data	-
Description of measurement methods and procedures applied	The temperature of the gas is required to determine the density of the methane combusted. The project employs landfill gas flow meter to measures flow, pressure and temperature and the normalized flow of landfill gas is displayed.
Frequency of monitoring/recording	-
Value applied:	-
Monitoring equipment	-

QA/QC procedures applied	Operators must receive training to master skills in equipment installation, maintenance, and troubleshooting, avoiding deviations caused by human operational errors.
Purpose of data	Calculation of baseline emissions
Calculation method	-
Comments	If the landfill gas flow meter employed measures flow, pressure and temperature and displays or outputs the normalized flow of landfill gas, then there is no need for separate monitoring of pressure and temperature of the landfill gas.

Data / Parameter	EE _y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	-
Description of measurement methods and procedures applied	As per paragraph 21 of AMS.III.G. (Version 10.0) , There are two options to decide the value of EE _y . Option 1: Specification provided by the equipment manufacturer. The equipment shall be designed to utilize biogas as fuel, and the efficiency specification is for biogas. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation. Option 2: Default efficiency of 40 percent. Default value 40% is applied in the project.
Frequency of monitoring/recording	-
Value applied:	40%
Monitoring equipment	-
QA/QC procedures applied	-
Purpose of data	Calculation of baseline emissions

Calculation method	-
Comments	-

5.3 Monitoring Plan

1. The requirement of monitoring plan

The project participants will monitor the emission reductions by methods, indicators, and frequency required by Monitoring Methodologies AMS-III.G. (Version 10.0) and AMS-I.D. (version 18.0) to ensure project ERs are measurable and real. The monitoring methodologies are based on direct measurement of the amount of landfill gas captured and destroyed by the project and electricity generating units.

2. Responsibilities of operational and management structure

The project participant will implement this monitoring plan. The plan could be revised according to suggestions from VVB and the practical circumstances, in order to keep it consistent, transparent and conservative during the monitoring process.

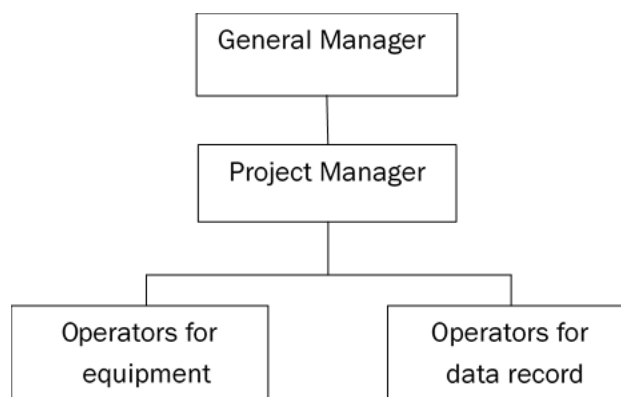


Figure 5.1 Operational and management structure

(1) Principal of the monitoring procedure

The general manager of the project is the leader of the monitoring tasks: he sets out the responsibility of everyone in the monitoring system, and establishes the related documents. The general manager ensures that staff in the monitoring system has the ability to deal with the assigned tasks.

(2) Executive person of the monitoring procedure

Project Manager: a Project Manager is appointed, specifically responsible for training, checking the daily operation, reporting forms and archiving emergency situation reports. The Project Manager reports monthly to the General Manager (GM) about the project performance and monitored data. In the event that non-conformance in the performance to the mentioned

procedures and/or functioning problems of the monitoring equipment are identified, the Project Manager will inform the GM about the situation and work out relevant measures to be taken. The Project Manager will also be responsible for aggregating the monitored data monthly and yearly, archiving and keeping data during the crediting period and two years after.

(3) Operators of the monitoring procedure

Operators will take turns to work in the control center 24 hours a day. They will be in charge of data supervision, filling operation report forms and, checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency situation reports.

3. Monitoring system

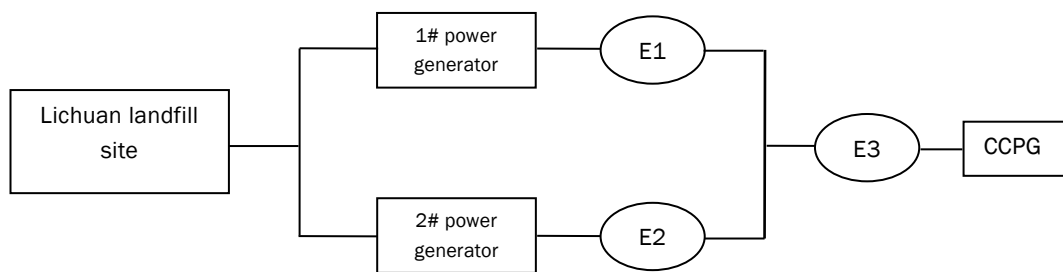


Figure 6.2 Monitoring system

Main parameters:

E1	Electricity meter to continuously measure the electricity generation by 1# power generator
E2	Electricity meter to continuously measure the electricity generation by 2# power generator
E3	Electricity meter (main meter, bi-directional) to continuously measure the electricity supplied to the CCPG and the electricity supplied by the grid

Data collection

The data are fully recorded and archived by DCS automatically and shown in the control system. The data are recorded once per hour. DCS of a historian database, primarily to provide data backup to the sheet of excel. Ultimately it forms the Monitoring Operating Record. All data are kept 2 years after the end of the crediting period.

Internal auditing and Quality assurance

Internal Audit System has been established for the project, and internal audit is carried out regularly according to the system requirements.

All metering equipment for monitoring will be chosen in accordance with VCS requirements, and will be calibrated regularly for accuracy by qualified party according to the national regulations. To assist in future verifications, the Project proponent will preserve the calibration records, along with the data files of project monitoring.

All the meters will be checked and maintained periodically.

Emergency procedures

Once a meter in fault, it shall be immediately repaired or replaced with another calibrated meter by a professional engineer, and the LFG or electricity generated during the period of erroneous measurement and replacement of the fault meter shall not be accounted for conservative consideration.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

The project does not involve commercially sensitive information.

Section	Information	Justification
/	/	/